

A Mathematical Representation of Near-Fault Ground Motions

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Abstract A simple, yet effective, analytical model is proposed for the representation of near-field strong ground motions. The model adequately describes the impulsive character of near-fault ground motions both qualitatively and quantitatively. In addition, it can be used to analytically reproduce empirical observations that are based on available near-source records. The input parameters of the model have an unambiguous physical meaning. The proposed analytical model has been calibrated using a large number of actual near-field ground-motion records. It successfully simulates the entire set of available near-fault displacement, velocity, and (in many cases) acceleration time histories, as well as the corresponding deformation, velocity, and acceleration response spectra. Furthermore, a very simplified methodology for generating realistic synthetic ground motions that are adequate for engineering analysis and design is outlined and applied. Finally, it should be noted that the analytical model (along with the scaling laws of its parameters) proposed in the present work has the potential to facilitate the study of the elastic and inelastic response of conventional, nonconventional (e.g., base-isolated), and special structures (e.g., suspension bridges, fluid-storage tanks) subjected to near-source seismic excitations as a function of the model input parameters and thus, ultimately, as a function of earthquake size.

Introduction

The characterization, proper parameterization, analytical modeling, and numerical simulation of near-fault seismic ground motions, as well as the study of their effects on the performance of engineering structures, are currently active research topics for both seismologists and earthquake engineers. The impetus has been provided by the devastating effect that recent seismic events (such as the 1994 Northridge, California, the 1995 Kobe, Japan, and the 1999 Izmit, Turkey, earthquakes) had on urban infrastructure. For all of these events, the causative fault was in the immediate vicinity of large metropolitan centers.

Two Californian earthquake events may be characterized as historical milestones related to near-source ground motions: the 1966 Parkfield and the 1971 San Fernando earthquakes.

The 1966 Parkfield, California, event provided the now famous Station 2 (C02) record at a distance of only 80 m from the fault break (Housner and Trifunac, 1967). Modern quantitative analysis of strong ground motion observations started with this record. Aki (1968) and Haskell (1969) demonstrated that the observed transverse (i.e., fault-normal) displacement component of this ground-motion record, which exhibited a simple impulsive form, was precisely what is expected for a right-lateral strike-slip rupture propagating from northwest to southeast (Fig. 1a).

The 1971 San Fernando, California, earthquake pro-

vided the equally well-known Pacoima Dam (PCD) record (Fig. 1b). The strike-normal velocity component of this record also exhibited an impulsive character that several investigators attempted to model (e.g., Boore and Zoback, 1974; Niazy, 1975; Bouchon, 1978b). In addition, this record was the one that made earthquake engineers recognize the severe implications of the impulsive characteristics of near-source ground motions on flexible structures. Bertero *et al.* (1978), with their study of the damage sustained by the Olive View Hospital during this event, were the first earthquake engineers to point this out. Since then, the occurrence of well-recorded major events has confirmed the existence of these directivity pulses in near-fault regions, as well as their destructive potential when the causative fault is in the immediate vicinity of large metropolitan areas.

The gradually increasing number of recorded near-source time histories has recently enabled strong motion seismologists to analyze more precisely the character of the near-fault ground motions and therefore contribute to the physical understanding of those features that control them (e.g., Campillo *et al.*, 1989; Somerville and Graves, 1993; Abrahamson and Somerville, 1996; Somerville *et al.*, 1997; O'Connell, 1998; Oglesby *et al.*, 1998; Papageorgiou, 1998; Somerville, 1998, 2000; O'Connell and Ake, 2002). Mavroeidis and Papageorgiou (2002) presented a comprehensive review and study of the factors that influence the near-source

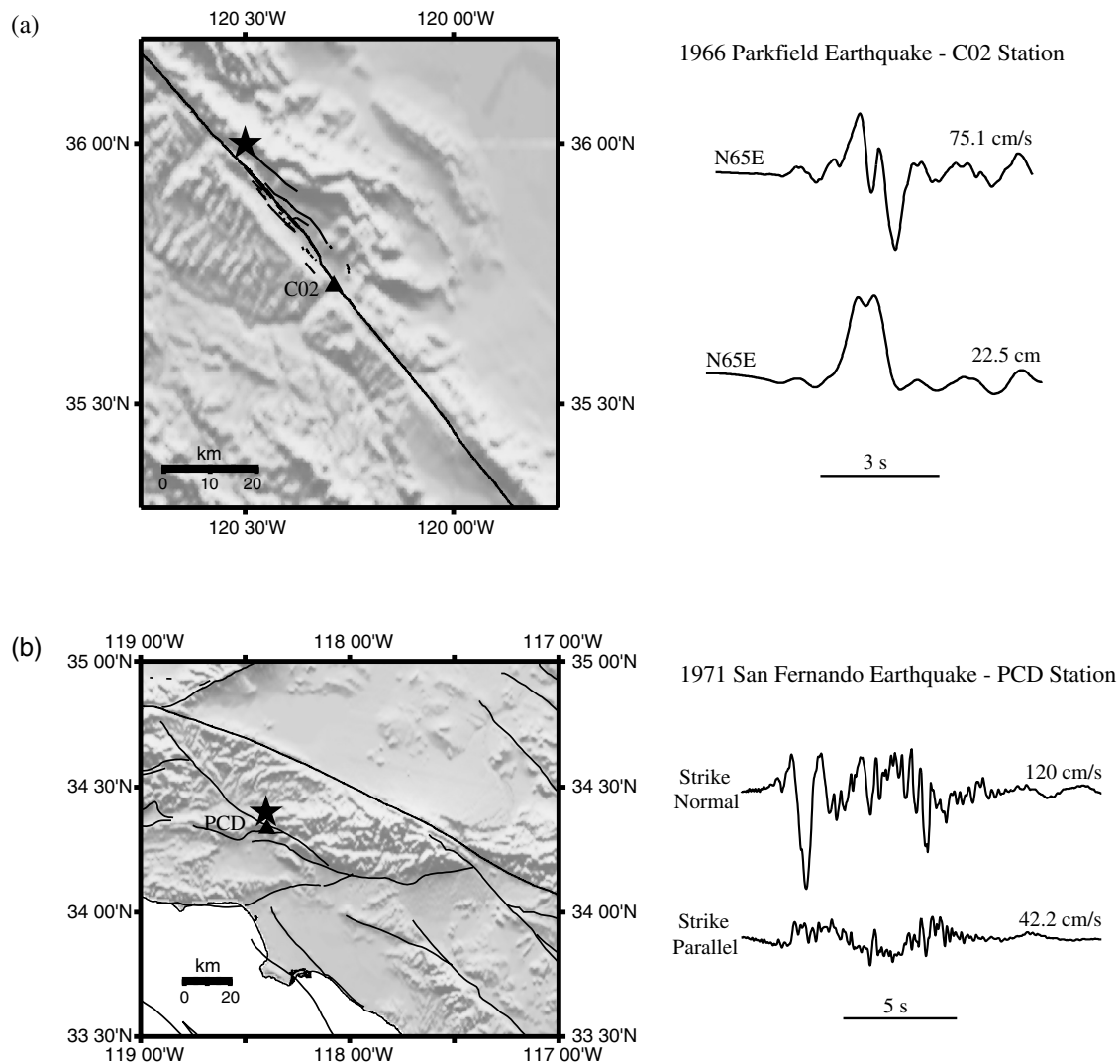


Figure 1. Characteristic near-fault strong ground motions: (a) Station 2 (C02) record obtained from the 1966 Parkfield, California, earthquake; (b) Pacoima Dam (PCD) record obtained from the 1971 San Fernando, California, earthquake.

ground motions. In that work, a large number of near-field records were used to examine the near-fault ground-motion characteristics pertaining to the rupture process. Even though emphasis was given on the forward directivity and permanent translation effects, other causes of the strong pulses observed in near-field regions were also addressed. In addition, a discussion regarding differences in ground motions generated by shear dislocation and crack-type ruptures was presented, an issue that is usually overlooked in practice.

We referred to Bertero *et al.* (1978) as the first instance of recognition of the special character of near-source ground motions from an earthquake engineering point of view. However, it was only after the 1994 Northridge, California, earthquake that the majority of engineers recognized the severe implications of such pulses on the performance of long-period structures and started considering methods to incor-

porate the near-source effects in engineering design. Code provisions have historically been developed based on recorded ground motions not sufficiently close to the causative fault. Anderson and Bertero (1987), Hall *et al.* (1995), Iwan (1997), Makris (1997), Chopra and Chintanapakdee (1998, 2001), Anderson *et al.* (1999), Kasalanati and Constantinou (1999), Malhotra (1999), Hall and Ryan (2000), Makris and Chang (2000a,b), Sasani and Bertero (2000), Alavi and Kra-winkler (2001), Cuesta and Aschheim (2001), Jangid and Kelly (2001), Liao *et al.* (2001), MacRae *et al.* (2001), Mylonakis and Reinhorn (2001), Rao and Jangid (2001), and Zhang and Iwan (2002a,b) have experimentally or analytically studied the elastic or inelastic response of engineering structures subjected to actual near-fault records or simplified waveforms intending to represent the typical ground-motion pulses observed in near-field regions.

Despite the progress that has been accomplished so far

by seismologists and engineers, there is still a clear need to develop simple, yet reliable, analytical models capable of providing synthetic motions that capture the impulsive character of near-source recordings both qualitatively and quantitatively. The model input parameters should have a clear physical meaning and be directly, to the extent possible, related to physical parameters of the faulting and wave-propagation processes. Furthermore, such a development would enable engineers to study the response of conventional, nonconventional (e.g., base-isolated), and special structures (e.g., fluid-storage tanks, suspension bridges) to near-fault seismic excitations more effectively as a function of the input parameters of the analytical models and thus, ultimately, of the earthquake size. The study of the dynamic structural response using simplified velocity waveforms (e.g., square, triangular, or sinusoidal pulses), although in the right direction, may yield misleading conclusions if the employed pulses do not capture the time history and response spectrum characteristics of the actual near-source records.

In this article, we present a simple mathematical model that serves the aforementioned purposes. The proposed model has been calibrated using a large number of actual near-fault ground-motion records. It successfully simulates the entire set of available displacement, velocity, and (in many cases) acceleration time histories, as well as the corresponding deformation, velocity, and acceleration response spectra (e.g., see Chopra [2001] for a definition and derivation of response spectra). An “objective” definition of the pulse duration is given based on model input parameters. In addition, we investigate the scaling characteristics of the model parameters with earthquake magnitude. Also, we derive the Fourier transform of the analytical model and identify the parameters that have the most significant effect on the spectral characteristics of the model. Finally, a simplified (yet adequate for engineering purposes) method is proposed for the synthesis of near-fault ground motions.

In another article (Mavroeidis *et al.*, 2003), we presented a comprehensive study of the elastic and inelastic response of single-degree-of-freedom (SDOF) systems subjected to synthetic near-fault ground motions generated by the analytical model proposed in the present work.

Database and Characteristics of Near-Source Strong Ground Motions

The near-fault strong ground motion database that we have compiled consists of a large number of processed near-field strong ground motion records from a variety of tectonic environments worldwide, including those generated by the recent 1999 Izmit, Turkey, 1999 Chi-Chi, Taiwan, and 1999 Duzce, Turkey, earthquakes. A database of 165 recorded ground-motion time histories, from different fault types (i.e., strike-slip, reverse, oblique, normal) and earthquake magnitudes (i.e., M_w 5.6–8.1), has been compiled from well-known and extensively studied seismic events that have occurred in the United States, Canada, Mexico, Japan, Greece,

Turkey, Romania, the former USSR, Iran, India, and Taiwan. In the database, we have also included five records obtained from fault ruptures initiated at large hypocentral depths (Olympia, Washington, 1949; Puget Sound, Washington, 1965; Bucharest, Romania, 1977) because they exhibit the characteristic impulsive character of near-source records. With the exception of the five records mentioned, all other motions were recorded at stations located within 20 km from the causative fault. A comprehensive spreadsheet of the database (including references) is provided in the Appendix so that other researchers may benefit from our work.

Although all stations (with the exception of the five deep-focus records/stations mentioned) included in the database are located in the vicinity of their respective causative faults, less than half of the database recordings exhibit strong velocity pulses. The near-source velocity time histories that exhibit “distinct” pulses are listed in Table 1. These data serve as a basis for the calibration of the analytical model presented herein. All of the records of Table 1 manifest forward directivity, except for five (YPT, SKR, GBZ^(SP), TCU052, TCU068), which have been identified to include the permanent translation effect. Information regarding site conditions, closest distance to the fault, and peak ground velocity values is also provided in Table 1.

In general, forward directivity and permanent translation are the two main causes for the velocity pulses observed in near-field regions (Abrahamson, 2000), even though other conditions (e.g., surface *P*-wave [e.g., Kawase and Aki, 1990], supershear rupture velocity [e.g., Bouchon *et al.*, 2001], or special geometrical conditions [e.g., Oglesby and Archuleta, 1997]) may also give rise or further intensify velocity pulses (Mavroeidis and Papageorgiou, 2002). *Forward directivity* occurs when the fault rupture propagates toward a site with a rupture velocity approximately equal to the shear-wave velocity. In this case, most of the elastic energy arrives coherently in a single, intense, relatively long-period pulse at the beginning of the record, representing the cumulative effect of almost all the seismic radiation from the fault. The phenomenon is even more pronounced when the direction of slip on the fault plane points toward the site as well (e.g., Somerville, 2000). On the other hand, *permanent translation* at a site is a consequence of permanent fault displacement due to an earthquake; it appears in the form of step displacement and one-sided velocity pulse in the strike-parallel direction for strike-slip faults (e.g., stations YPT and SKR from the 1999 Izmit, Turkey, earthquake) or in the strike-normal direction for dip-slips faults (e.g., stations TCU052 and TCU068 from the 1999 Chi-Chi, Taiwan, earthquake). In the latter case, directivity and translation effects “build up” in the same direction and, for analysis purposes, should be decoupled.

Figure 2 illustrates the velocity pulses of all records listed in Table 1 in a visually informative manner. These ground-motion records are oriented in the strike-normal and strike-parallel directions. However, for seismic events characterized by uncertainties regarding the strike direction of

Table 1
Near-Fault Records with Distinct Velocity Pulses for the Calibration of the Analytical Model

No.	Location	Date	Fault Mechanism*	M_w	Station	Site Code†	Closest Fault Distance (km)	Component‡	Recorded PGV (cm/sec)	Effect§
1	Parkfield, CA, USA	27-Jun-66	SS	6.20	C02	SL	0.1	SN	75.1	FD
2	San Fernando, CA, USA	09-Feb-71	RV	6.55	PCD	HR	3.0	SN	120.0	FD
3	Gazli, USSR	17-May-76	RV	6.80	KAR	SR	3.0	Rad	60.0	FD
4	Bucharest, Romania	04-Mar-77	RV	7.27	BRI	SL	190.0	SN	74.9	FD
5	Tabas, Iran	16-Sep-78	RV	7.11	TAB	SL	1.2	SP	122.0	FD
6	Coyote Lake, CA, USA	8-Jun-79	SS	5.63	GA6	SR	1.2	SN	47.5	FD
7	Imperial Valley, CA, USA	15-Oct-79	SS	6.50	E04	SL	6.0	SN	78.3	FD
					E05	SL	2.7	SN	91.8	FD
					E06	SL	0.3	SN	112.0	FD
					E07	SL	1.8	SN	109.0	FD
					EMO	SL	1.2	SN	115.0	FD
8	Mexicali Valley, Mexico	9-Jun-80	SS	6.37	VCT	SL	3.0	SN	76.7	FD
9	Morgan Hill, CA, USA	24-Apr-84	SS	6.15	HAL	SL	2.0	SN	39.8	FD
10	Palm Springs, CA, USA	8-Jul-86	OB	6.09	NPS	SL or SR	4.0	SN	73.6	FD
					DSP	SL	6.4	SN	29.2	FD
11	Whittier Narrows, CA, USA	10-Oct-87	RV	5.93	DOW	SL	16.4	SN	30.7	FD
					NWK	SL	15.7	SN	20.0	FD
12	Superstition Hills, CA, USA	24-Nov-87	SS	6.40	PTS	?	0.7	SN	109.0	FD
					ELC	SL	13.6	SN	52.0	FD
13	Loma Prieta, CA, USA	17-Oct-89	OB	6.90	LGP	HR	3.0	SN	102.0	FD
					STG	SL	8.3	SN	56.4	FD
14	Sierra Madre, CA, USA	28-Jun-91	RV	5.56	COG	?	9.4	Rad (filt)	15.3	FD
15	Erzincan, Turkey	13-Mar-92	SS	6.63	ERZ	SL	2.0	SN	95.2	FD
16	Landers, CA, USA	28-Jun-92	SS	7.20	LUC	SL	1.1	SN	114.0	FD
17	Northridge, CA, USA	17-Jan-94	RV	6.70	JFA	SL	5.2	SN	105.0	FD
					RRS	SL	6.0	SN	173.0	FD
					SCG	SL	5.1	SN	134.0	FD
					SCH	SL	5.0	SN	122.0	FD
					NWS	SL	5.3	SN	117.0	FD
18	Aigion, Greece	15-Jun-95	NM	6.33	AEG	SL or SR	6.0	Long	40.9	FD
					AEG	SL or SR	6.0	Tran	52.0	FD
19	Izmit, Turkey	17-Aug-99	SS	7.40	ARC	SR	14.0	SN	44.3	FD
					SKR	SR	3.1	SP	80.3	PT
					YPT	SL	2.6	SP	87.8	PT
					GBZ	SR	11.0	SN	41.4	FD
					GBZ	SR	11.0	SP	28.7	PT
20	Chi-Chi, Taiwan	20-Sep-99	OB	7.60	TCU052	SL	0.8	SN	270.0	PT
					TCU068	SL	0.2	SN	380.0	PT
					TCU075	SL	0.6	SN	115.0	FD
					TCU076	SL	2.3	SN	88.0	FD
					TCU129	SL	1.5	SN	68.0	FD

*SS, strike-slip; RV, reverse; OB, oblique; NM, normal.

†HR, hard rock; SR, sedimentary and conglomerate rock; SL, soil and alluvium.

‡SN, strike-normal; SP, strike-parallel; Rad, radial; Long, longitudinal; Tran, transverse.

§FD, forward directivity; PT, permanent translation.

the causative fault plane (KAR record) and for near-fault records characterized by uncertainties regarding the precise orientation of the recorded horizontal ground-motion components (AEG record), we have decided either to rotate the original strong motion records in the radial/transverse direction (with respect to the epicenter) or present them in the original directions provided in the data files released by the agencies. These few records, although not oriented in the standard strike-normal and strike-parallel directions, are still characterized by distinct velocity pulses. By inspecting the signals of Figure 2, it becomes apparent that many of them

look alike. It can be stated that rupture directivity produces a narrowband velocity pulse, the period of which increases with earthquake magnitude (for a comprehensive discussion regarding the characteristics of near-source ground motions see, e.g., Mavroeidis and Papageorgiou [2002]).

A Simple Analytical Model for Near-Fault Pulses

The *pulse duration* (or *period*), the *pulse amplitude*, as well as the *number* and *phase of half cycles* are the key parameters that define the waveform characteristics of near-

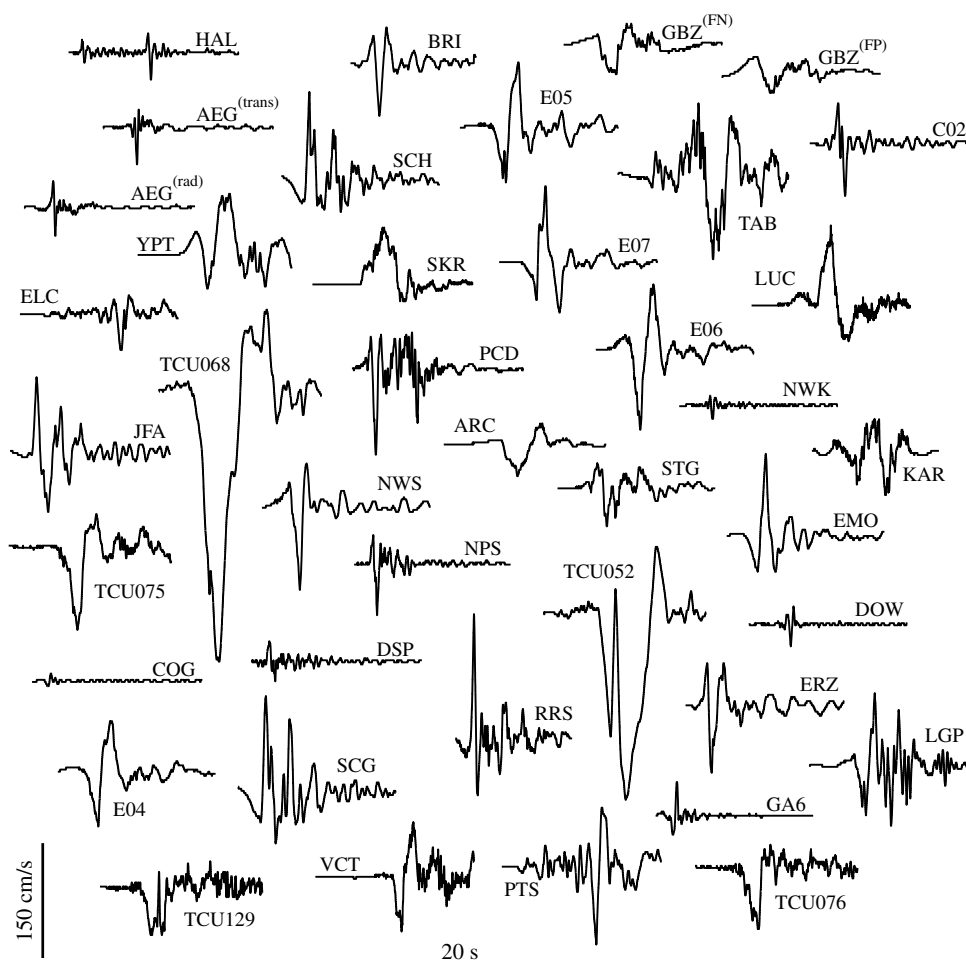


Figure 2. Strong ground motion records with distinct velocity pulses utilized for the calibration of the proposed analytical model (see also Table 1).

fault velocity pulses. Therefore, an analytical model with four parameters in principle should suffice to describe the entire set of velocity pulses generated due to forward directivity or permanent translation effects.

Seismologists have used “wavelets” (also referred to as “signals,” “signatures,” or “pulses”), particularly in fields such as seismic filtering, wavelet processing, wave-propagation modeling, and trace inversion (Hubral and Tygel, 1989). Although various wavelets have been proposed in the literature, only a limited number of them are popular and frequently used in practice. The most common wavelets are summarized in Table 2 along with their analytical expressions, input parameters, and associated references.

The analytical model of choice should satisfy the conditions and possess the following properties:

1. The synthetic wavelet should be expressed by a properly parameterized simple mathematical expression that, with a minimum number of input parameters that have an unambiguous physical interpretation, allows as much flexibility as is necessary to represent, reasonably accurately, near-source pulses.

2. The synthetic wavelet should be capable of simulating as many as possible of the near-source records (preferably all of the records in Table 1).
3. The mathematical expression of the wavelet should be such that it facilitates derivation of closed-form expressions of its spectral characteristics in the form of Fourier transform and response spectra. Such closed-form expressions make considerably easier the parametric study of the response of structures to near-source pulses.

From the wavelets listed in Table 2, the analytical model that fulfills most of the aforementioned conditions and has the potential, if slightly modified, to entirely meet our requirements is the Gabor wavelet. This signal is the product of a harmonic oscillation and a bell-shaped function (i.e., Gaussian envelope). The Gabor wavelet is defined by the four parameters identified at the beginning of this section as the key features that determine the waveform characteristics of the near-source velocity pulses; namely, A , f_p , ν , and γ (see Table 2) define the amplitude, prevailing frequency, phase, and oscillatory character of the signal, respectively.

The Berlage wavelet is similar to the Gabor signal be-

Table 2
Common Analytic Wavelets Used in Seismology

Wavelet	Analytical Expression	Parameters	References
Gabor	$f(t) = Ae^{-(2\pi f_p/\gamma)^2 t^2} \cos[2\pi f_p t + \nu]$	A: amplitude f_p : prevailing frequency ν : phase angle γ : oscillatory character	Gabor (1946) Morlet <i>et al.</i> (1982)
Berlage	$f(t) = AH(t)t^n e^{-(2\pi f_p/\gamma)t} \cos[2\pi f_p t + \nu]$	A: amplitude f_p : prevailing frequency ν : phase angle γ : oscillatory character n : asymmetry of envelope function Note: $H(t)$ is the Heaviside unit step function	Berlage (1932) Kulhánek and Klíma (1970) Farnbach (1975) Aldridge (1990)
Generalized Rayleigh	$f(t) = A(-1)^k \frac{e^{i(\nu + \pi/2)}}{\left(i + \frac{2\pi f_p t}{k}\right)^{k+1}}$ Use: $f_R(t) = \text{Re}\{f(t)\}$ or $f_I(t) = -\text{Im}\{f(t)\}$	A: amplitude f_p : prevailing frequency ν : phase angle k : controls the number of “lobes”	Hudson, 1980 Hubral and Tygel (1989)
Küpper	$f(t) = A \left[\sin\left(m \frac{\pi t}{T}\right) - \frac{m}{m+2} \sin\left((m+2) \frac{\pi t}{T}\right) \right]$ for $0 < t < T$	A: amplitude T : duration m : controls the number of half-cycles	Küpper (1958) Müller (1970) Fuchs and Müller (1971)
Ricker	Three-loop: (symmetric) $f(t) = A(1 - 2\pi^2 f_p^2 t^2) e^{-(\pi f_p)^2 t^2}$ Two-loop: (antisymmetric) $f(t) = A t e^{-(\sqrt{2}\pi f_p)^2 t^2}$	A: amplitude f_p : prevailing frequency A: amplitude f_p : prevailing frequency	Ricker (1943) Ricker (1944) Ricker (1945) Ricker (1953a) Ricker (1953b) Dietrich and Bouchon (1985) Hosken (1988)

cause both of them consist of an amplitude-modulated harmonic. Furthermore, the common input parameters of the two signals have the same physical meaning. However, the Berlage wavelet is characterized by an additional free parameter (parameter n ; see Table 2) that controls the skewness of its envelope function. This extra degree of freedom, although theoretically useful, is not of great practical importance. The need for a skewed (i.e., nonsymmetric) envelope may be accommodated (to a certain extent) by varying the phase angle of the amplitude-modulated harmonic. Furthermore, the existence of the factor t^n in the mathematical expression of the Berlage wavelet introduces additional complexity in the analytical derivations to follow, without significant benefits.

The three-loop symmetric and two-loop antisymmetric Ricker wavelets are not adequate to describe a broad range of near-fault velocity pulses; only a small number of recorded ground motions exhibit purely symmetrical or antisymmetrical characteristics, as is evident from Figure 2. There is no unique mathematical expression for “arbitrary” (meaning not perfectly symmetrical or antisymmetrical) Ricker wavelets; depending on the waveform to be obtained, different polynomial functions should be selected (Ricker, 1944, 1945). Hosken (1988) discussed the applicability and

usefulness of the Ricker wavelet, focusing on the drawbacks of the signal. The author also proposed a technique to generalize the Ricker wavelet by generating signals in between the antisymmetric two-loop and the symmetric three-loop Ricker wavelets. Besides the complexity in the mathematical expressions of the generalized Ricker wavelets, it is apparent that a two-parameter model is far too constrained to allow accurate fitting of the fairly complicated recorded near-fault velocity pulses.

Like the Ricker wavelet, the original Rayleigh signal (e.g., Hudson, 1980) is a two-parameter model inflexible for most synthetic seismogram applications. Hubral and Tygel (1989) overcame this deficiency by deriving a generalized analytic wavelet based on the Rayleigh signal. The real or imaginary part of the mathematical expression of this generalized wavelet may be used to generate synthetic velocity pulses. Even though this is not a difficult task, the resulting formulas are quite complicated for further derivations (e.g., differentiation and integration to obtain acceleration and displacement time histories, respectively; response of an SDOF system; etc.).

On the other hand, the analytical expression of the Küpper wavelet is straightforward and its input parameters have a clear physical meaning. However, the waveforms gener-

ated by the Küpper signal are either symmetric or antisymmetric for odd or even m values, respectively. The lack of a phase parameter does not facilitate generation of “arbitrary” signals. This is an important limitation previously addressed with respect to the Ricker wavelet.

Based on the above discussion, the Gabor wavelet is the analytical model (among those listed in Table 2) that better serves our needs. However, no closed-form solution can be derived for the response of an SDOF system when excited by synthetic ground motions generated by the Gabor signal. Such a closed-form expression would greatly facilitate parametric analyses, a matter of great importance to earthquake engineers. The difficulties in the derivations are caused by the exponential function (i.e., Gaussian envelope) included in the mathematical expression of the Gabor wavelet. Therefore, to overcome this difficulty, we propose an analytical model that retains the advantages of the Gabor wavelet (e.g., number of parameters, physical interpretation of them, simple mathematical expression, large flexibility in synthetic waveforms), while at the same time yields a closed-form expression for the response of the SDOF system subjected to synthetic ground motions generated by the model. To that effect, we have replaced the Gaussian envelope of the Gabor wavelet by another symmetric bell-shaped function that possesses a simpler analytical expression. Namely, a shifted haversed sine function (i.e., an elevated cosine function) is used to replace the Gaussian envelope, while the harmonic oscillation part remains the same. Thus, the proposed analytical signal is expressed by

$$f(t) = A \frac{1}{2} \left[1 + \cos\left(\frac{2\pi f_p}{\gamma} t\right) \right] \cos(2\pi f_p t + \nu). \quad (1)$$

The following remarks can be made pertaining to the proposed analytical model:

- The shifted haversed sine function is a periodic function; consequently, it does not produce an envelope with a single hump like the Gaussian function of the Gabor wavelet. This problem is easily resolved by limiting the time interval of the signal as follows:

$$-\frac{\gamma}{2f_p} \leq t \leq \frac{\gamma}{2f_p}. \quad (2a)$$

- The period of the harmonic oscillation should be smaller than the period of the envelope represented by the elevated cosine function in order to produce physically acceptable signals; that is,

$$\frac{1}{f_p} < \frac{\gamma}{f_p} \Rightarrow \gamma > 1. \quad (2b)$$

- It is convenient for the calibration of the model to introduce a time shift, t_0 , in equation (1) to precisely define the epoch of the envelope's peak. This parameter is frequently

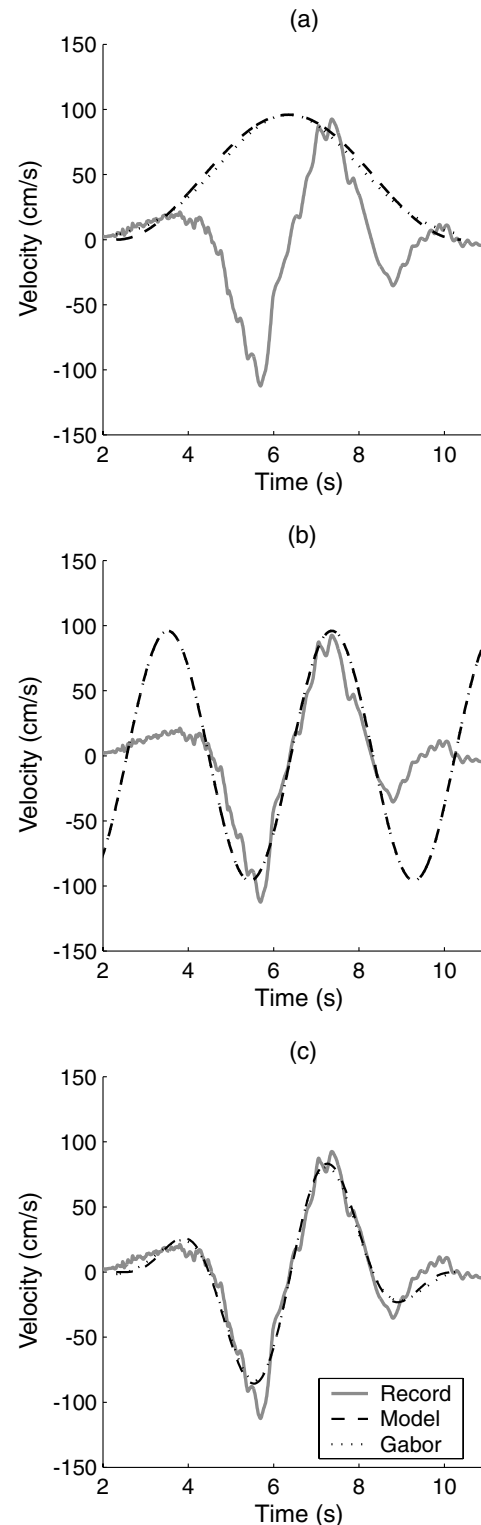


Figure 3. Comparison of envelopes, carriers and pulses generated by the proposed analytical model (i.e., $A = 96$, $\nu = 265^\circ$, $f_p = 0.26$, and $\gamma = 2.1$) and the Gabor wavelet (i.e., $A = 96$, $\nu = 265^\circ$, $f_p = 0.26$, and $\gamma = 3.7$) simulating the fault-normal velocity component recorded at station E06 during the 1979 Imperial Valley, California, earthquake. (a) Envelopes; (b) carriers; (c) pulses.

introduced in all models listed in Table 2 as a feature that provides extra flexibility to the signal, allowing its translation along the time axis. Thus,

$$t \Rightarrow t - t_0. \quad (2c)$$

The combination of equations (1) and (2) yields the formulation of the proposed analytical model for the near-fault ground velocity pulses:

$$v(t) = \begin{cases} A \frac{1}{2} \left[1 + \cos\left(\frac{2\pi f_p}{\gamma} (t - t_0)\right) \right] \cos[2\pi f_p(t - t_0) + \nu], \\ 0, \text{ otherwise} \end{cases} \quad (3)$$

$$t_0 - \frac{\gamma}{2f_p} \leq t \leq t_0 + \frac{\gamma}{2f_p} \quad \text{with } \gamma > 1.$$

Parameter A controls the amplitude of the signal, f_p is the frequency of the amplitude-modulated harmonic (or the prevailing frequency of the signal), ν is the phase of the amplitude-modulated harmonic (i.e., $\nu = 0$ and $\nu = \pm\pi/2$ define symmetric and antisymmetric signals, respectively), γ is a parameter that defines the oscillatory character (i.e., zero crossings) of the signal (i.e., for small γ the signal approaches a deltalike pulse; as γ increases the number of zero crossings increases), and t_0 specifies the epoch of the envelope's peak.

To demonstrate that the proposed mathematical model produces almost identical pulses with the Gabor wavelet and

therefore justify the substitution of the Gaussian envelope with the elevated cosine function, both analytical models have been used to generate synthetic signals that simulate the fault-normal velocity pulse recorded at station E06 during the 1979 Imperial Valley, California, earthquake. The comparison is illustrated in Figure 3; the harmonic oscillation (i.e., the carrier signal) is the same for both pulses, while their envelope functions are very similar. The synthetic signals produced by both models reproduce the longer-period portions of the observed ground motion.

A significant feature of the proposed analytical model is the objective definition of the pulse duration based on model input parameters. In the literature, no unique method exists to define the duration or period of the velocity pulse, even though this parameter is extensively used. Many researchers determine the pulse duration using the zero crossings of the pulse waveform. Others prefer the utilization of the peak value of the velocity response spectrum to indirectly define the pulse period. In many other cases, no explanation is provided regarding the estimation of this parameter. We propose an objective definition of the pulse duration (T_p) compatible with the physical aspects of the problem as the inverse of the prevailing frequency (f_p) of the signal; that is,

$$T_p = \frac{1}{f_p}. \quad (4)$$

The analytical expressions for the ground acceleration and displacement time histories compatible with the ground velocity given by equation (3) are

$$a(t) = \begin{cases} -\frac{A\pi f_p}{\gamma} \left[\sin\left(\frac{2\pi f_p}{\gamma} (t - t_0)\right) \cos[2\pi f_p(t - t_0) + \nu] + \gamma \sin[2\pi f_p(t - t_0) + \nu] \left[1 + \cos\left(\frac{2\pi f_p}{\gamma} (t - t_0)\right) \right] \right], \\ 0, \text{ otherwise} \end{cases} \quad t_0 - \frac{\gamma}{2f_p} \leq t \leq t_0 + \frac{\gamma}{2f_p} \text{ with } \gamma > 1, \quad (5)$$

$$d(t) = \begin{cases} \frac{A}{4\pi f_p} \left[\sin[2\pi f_p(t - t_0) + \nu] + \frac{1}{2} \frac{\gamma}{\gamma - 1} \sin\left[\frac{2\pi f_p(\gamma - 1)}{\gamma} (t - t_0) + \nu\right] \right] + C, \\ \frac{A}{4\pi f_p} \left[\sin\left[\frac{2\pi f_p(\gamma + 1)}{\gamma} (t - t_0) + \nu\right] + \frac{1}{2} \frac{\gamma}{\gamma + 1} \sin[2\pi f_p(t - t_0) + \nu] \right] + C, \\ \frac{A}{4\pi f_p} \frac{1}{(1 - \gamma^2)} \sin(\nu - \pi\gamma) + C, \quad t < t_0 - \frac{\gamma}{2f_p} \\ \frac{A}{4\pi f_p} \frac{1}{(1 - \gamma^2)} \sin(\nu + \pi\gamma) + C, \quad t > t_0 + \frac{\gamma}{2f_p} \end{cases} \quad t_0 - \frac{\gamma}{2f_p} \leq t \leq t_0 + \frac{\gamma}{2f_p} \quad \gamma > 1. \quad (6)$$

The constant displacement values for $t < t_0 - \gamma/2f_p$ and $t > t_0 + \gamma/2f_p$ were specified so that the displacement time histories satisfy continuity condition at $t = t_0 - \gamma/2f_p$ and $t_0 + \gamma/2f_p$.

When the velocity pulse is integrated to obtain the closed-form displacement, an as yet unspecified constant, C , appears in the derived displacement expression, that is, $d(t)$

$= \int v(t)dt + C$ or $d(t) = d^I(t) + C$, where $d^I(t)$ represents the indefinite integral $\int v(t)dt$. In order to specify the above constant of integration, there are three options: (1) One may suggest imposing zero displacement for $t < t_0 - \gamma/2f_p$ and thus constrain the above integration constant, that is, $C = -d^I(t_0 - \gamma/2f_p)$. This is a reasonable hypothesis as long as the recorded displacement being simulated is approximately

zero for $t = t_0 - \gamma/2f_p$. (2) Another more complicated approach is to integrate the recorded velocity time history for $t < t_0 - \gamma/2f_p$ and require the corresponding displacement offset to be equal to the integration output. This may be a more accurate procedure, especially when wiggles or emergent-type time histories exist at the beginning of the velocity record that cannot be successfully simulated by the proposed analytical model, but yet lead to an accumulated displacement at $t = t_0 - \gamma/2f_p$. (3) Alternatively, we may assume that the integration constant is zero for the sake of simplicity, that is, $C = 0$. In such a case, there is a displacement offset for $t < t_0 - \gamma/2f_p$ equal to $d^1(t_0 - \gamma/2f_p)$ that may not be fully compatible with actual displacement records, even though there are displacement time histories that exhibit nonzero values for $t = 0$ due to the correction and processing schemes applied on the recorded digital data. However, as becomes apparent in the next section, where the model calibration is presented, this displacement offset is close to zero for the great majority of the examined cases. The few

records for which this offset is noticeable are characterized by velocity waveforms that are of the emergent type. On the other hand, the displacement offset for $t > t_0 + \gamma/2f_p$ [i.e., $d(t_0 + \gamma/2f_p)$] may reliably simulate the permanent (i.e., static) displacement related to the tectonic deformation experienced by the site due to the fault movement. In what follows, we adopt the third option, and we assume $C = 0$.

A parametric study in terms of ν and γ of the normalized (with respect to f_p and A) acceleration, velocity, and displacement pulses can be performed based on the equations presented previously. We define the *normalized time variable* as

$$\bar{t} = 2\pi f_p(t - t_0). \quad (7)$$

Then, the normalized acceleration, velocity, and displacement time histories can be expressed by rewriting equations (5), (3), and (6), respectively, as

$$\bar{a}(\bar{t}) = \frac{a(t)}{Af_p} = \begin{cases} -\frac{\pi}{\gamma} \left[\sin\left(\frac{\bar{t}}{\gamma}\right) \cos(\bar{t} + \nu) + \gamma \sin(\bar{t} + \nu) \left(1 + \cos\left(\frac{\bar{t}}{\gamma}\right)\right) \right], & -\pi\gamma \leq \bar{t} \leq \pi\gamma \text{ with } \gamma > 1, \\ 0, & \text{otherwise} \end{cases} \quad (8)$$

$$\bar{v}(\bar{t}) = \frac{v(t)}{A} = \begin{cases} \frac{1}{2} \left[1 + \cos\left(\frac{\bar{t}}{\gamma}\right) \right] \cos(\bar{t} + \nu), & -\pi\gamma \leq \bar{t} \leq \pi\gamma \text{ with } \gamma > 1, \\ 0, & \text{otherwise} \end{cases} \quad (9)$$

$$\bar{d}(\bar{t}) = \frac{d(t)}{(A/f_p)} = \begin{cases} \frac{1}{4\pi} \left[\sin(\bar{t} + \nu) + \frac{1}{2} \frac{\gamma}{\gamma - 1} \sin\left(\frac{\gamma - 1}{\gamma} \bar{t} + \nu\right) + \frac{1}{2} \frac{\gamma}{\gamma + 1} \sin\left(\frac{\gamma + 1}{\gamma} \bar{t} + \nu\right) \right], & -\pi\gamma \leq \bar{t} \leq \pi\gamma \\ \frac{1}{4\pi} \frac{1}{(1 - \gamma^2)} \sin(\nu - \pi\gamma), & \bar{t} < -\pi\gamma \\ \frac{1}{4\pi} \frac{1}{(1 - \gamma^2)} \sin(\nu + \pi\gamma), & \bar{t} > \pi\gamma \end{cases} \quad \gamma > 1. \quad (10)$$

Figure 4 illustrates the variation of the normalized acceleration, velocity, and displacement series as a function of the normalized time for selected ν and γ values. For $\nu = 0^\circ$, 180° we obtain symmetrical pulses with reversed polarities, while for $\nu = 90^\circ$ we obtain antisymmetric pulses. In addition, as γ increases more oscillations appear in the signal. It becomes apparent that A affects the amplitude of the synthetic time histories only, while f_p affects the timescale of all synthetic time histories, as well as the acceleration and displacement amplitudes; on the other hand, the velocity amplitude is independent of f_p . These are important remarks that enable us to interpret empirical observations pertaining to the intensity of the near-fault ground motions as explained in another section.

Model Calibration

The next step is the calibration of the proposed analytical model using the actual near-fault strong ground motion records listed in Table 1. We have simultaneously fitted our mathematical model to the displacement, velocity, and acceleration time histories, as well as to their corresponding elastic response spectra. The input parameters obtained by the calibration procedure are summarized in Table 3. It is apparent that the t_0 values listed in Table 3 have no physical significance, however, they have been posted for anyone who may want to reproduce our results.

Estimation of the parameters T_p , A , ν , and γ is not accomplished by any formal least-squares inversion, but by using a step-by-step procedure, as follows. The pulse dura-

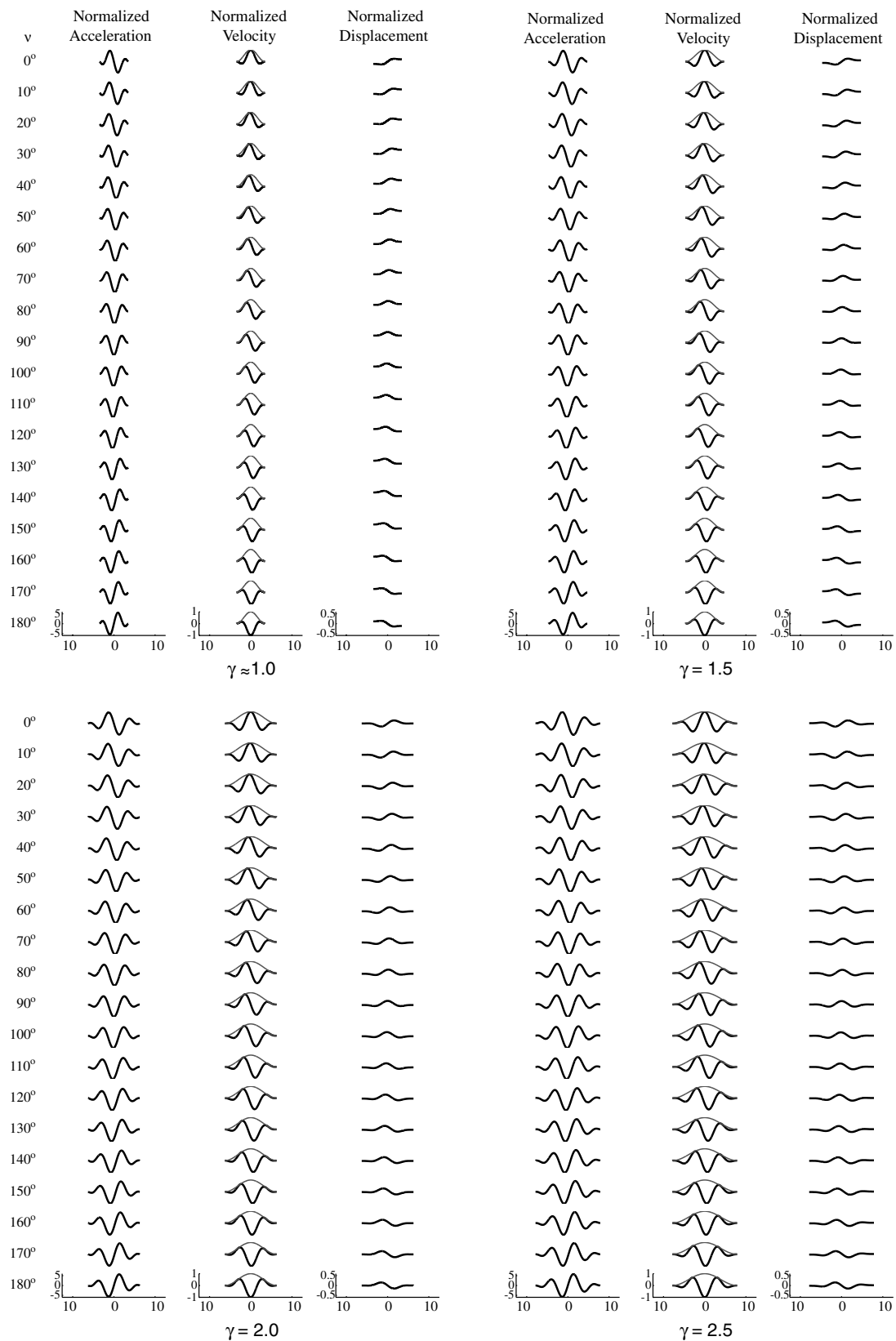


Figure 4. Normalized acceleration, velocity, and displacement pulses as function of the normalized time for a suite of ν and γ values. The gray lines over velocity pulses indicate the envelope functions.

Table 3
Model Input Parameters Obtained by Fitting the Analytical Model to Recorded Near-Fault Ground Motions

No.	Location	Station	Component*	A	γ	ν (°)	f_p (Hz)	t_0 (sec)
1	Parkfield, CA, USA	C02	SN	60.0	1.700	100.0	0.500	4.100
2	San Fernando, CA, USA	PCD	SN	115.0	1.600	180.0	0.680	3.000
3	Gazli, USSR	KAR	Rad	45.0	2.900	70.0	0.238	8.400
4	Bucharest, Romania	BRI	SN	62.0	2.400	200.0	0.470	3.830
5	Tabas, Iran	TAB	SP	104.0	2.200	180.0	0.190	12.400
6	Coyote Lake, CA, USA	GA6	SN	48.0	1.550	315.0	1.000	2.600
7	Imperial Valley, CA, USA	E04	SN	71.0	1.900	305.0	0.225	6.100
		E05	SN	84.0	1.900	300.0	0.255	6.600
		E06	SN	96.0	2.100	265.0	0.260	6.350
		E07	SN	79.0	2.100	25.0	0.275	6.100
		EMO	SN	78.0	2.300	0.0	0.340	4.950
8	Mexicali Valley, Mexico	VCT	SN	80.0	1.500	270.0	0.270	11.550
9	Morgan Hill, CA, USA	HAL	SN	38.0	1.750	130.0	1.150	10.450
10	Palm Springs, CA, USA	NPS	SN	60.0	1.700	170.0	0.800	2.890
		DSP	SN	21.0	2.000	80.0	0.600	2.520
11	Whittier Narrows, CA, USA	DOW	SN	29.0	2.200	220.0	1.200	5.350
		NWK	SN	20.0	2.100	200.0	1.350	4.300
12	Superstition Hills, CA, USA	PTS	SN	112.0	1.800	237.0	0.445	12.295
		ELC	SN	46.0	1.650	210.0	0.430	13.180
13	Loma Prieta, CA, USA	LGP	SN	60.0	3.000	280.0	0.310	7.600
		STG	SN	47.0	1.900	150.0	0.270	6.500
14	Sierra Madre, CA, USA	COG	Rad (filt)	9.2	2.300	260.0	1.100	2.190
15	Erzincan, Turkey	ERZ	SN	67.0	2.500	210.0	0.410	3.700
16	Landers, CA, USA	LUC	SN	100.0	1.210	55.0	0.170	10.500
17	Northridge, CA, USA	JFA	SN	87.0	2.300	100.0	0.330	4.120
		RRS	SN	142.0	1.700	20.0	0.800	2.430
		SCG	SN	93.0	2.500	0.0	0.340	3.650
		SCH	SN	80.0	2.300	0.0	0.330	3.770
		NWS	SN	94.0	1.700	200.0	0.370	4.770
18	Aigion, Greece	AEG	Long	44.5	1.450	75.0	1.400	3.840
		AEG	Tran	61.0	1.200	205.0	1.480	4.340
19	Izmit, Turkey	ARC	SN	41.0	1.380	225.0	0.140	14.700
		SKR	SP	67.0	1.023	5.0	0.105	9.000
		YPT	SP (first pulse)	35.0	1.550	90.0	0.190	9.150
			SP (second pulse)	91.5	1.050	10.0	0.137	12.050
		GBZ	SN	34.5	2.200	220.0	0.210	6.600
		GBZ	SP	28.0	1.800	85.0	0.165	4.900
20	Chi-Chi, Taiwan	TCU052	SN	225.0	1.048	190.0	0.079	35.400
		TCU068	SN	332.0	1.055	190.0	0.082	36.300
		TCU075	SN	105.0	1.200	220.0	0.170	28.500
		TCU076	SN	70.0	1.088	200.0	0.175	27.280
		TCU129	SN	52.0	1.090	200.0	0.130	27.100
21	Kobe, Japan [†]	KOB	SN (first pulse)	85.0	1.550	270.0	1.000	7.600
			SN (second pulse)	75.0	1.700	270.0	1.100	8.400
			SN (third pulse)	45.0	2.000	70.0	0.500	9.300

*Component abbreviations are as in Table 1.

[†]Not characterized by a distinct near-fault ground velocity pulse (i.e., thus, not included in Table 1).

tion, T_p , is determined so that the pseudo-velocity response spectra of the synthetic and recorded near-fault ground motions exhibit their peak values at approximately the same natural period. This does not necessarily mean that T_p coincides with the period corresponding to the peak of the pseudovelocity response spectrum of the actual near-fault record, even though this may hold true in many cases. The parameter A is determined so that the amplitude of the synthetic velocity pulse and its peak pseudospectral velocity agree well with the corresponding quantities of the actual

record. Finally, the parameters ν and γ are adjusted so as to optimize the fitting of the synthetic velocity and displacement records. It should be noted that the determination of parameters A , ν , and γ is performed simultaneously by trial and error since γ affects the pseudovelocity spectral amplitudes as well. Although one could introduce a formal least-squares fitting at various stages of the estimation process, we found such a formalism to be cumbersome, without any substantial benefits.

We could have selected to fit the analytical model to the

recorded velocity time histories only and subsequently use the estimated input parameters to generate the corresponding compatible synthetic acceleration and displacement time series, as well as the synthetic response spectra. However, although slightly different sets of input parameters may generate synthetic velocity pulses that fit, for all practical purposes, equally well the recorded ground velocity, not all sets produce synthetic displacements and response spectra that successfully replicate the recorded ones. Thus, the simultaneous fitting of time histories and response spectra appears to be a more accurate and reliable procedure for the calibration of the proposed analytical model.

Figure 5 displays many samples of the quality of fitting of the synthetic signal to the recorded strong ground motions listed in Table 1. It is evident that the proposed mathematical model successfully simulates the entire set of recorded near-fault displacement, velocity, and (in many cases) acceleration time histories both qualitatively and quantitatively. As was previously pointed out, the proposed analytical model is meant to replicate accurately the intermediate- to long-period features of ground motion. Consequently, the high-frequency components of ground motion that are stochastic in nature and are clearly evident in accelerograms cannot be simulated using the proposed analytical model.

The synthetic elastic response spectra are also in good agreement with the elastic response spectra of the recorded ground motions in the vicinity of the prevailing frequency, f_p , of the pulse. For moderate earthquakes, velocity pulses are characterized by relatively short duration and the agreement between synthetic and recorded elastic response spectra is very good over the entire frequency range. However, for large earthquakes, velocity pulses have a long duration and the fitting of the spectra is satisfactory only in the long-period range.

Actual near-fault velocity records may consist of more than one velocity pulse generated by distinct slip patches on the fault plane. In such cases, we can utilize a sequence of synthetic pulses to accurately fit the recorded ground motions. Typical examples of such a case are illustrated in Figure 6. This figure displays the strike-parallel component of the strong ground motion recorded at the Yarımcı station (YPT) during the 1999 İzmit, Turkey, earthquake and the strike-normal component of ground motion recorded at Kobe station (KOB) during the 1995 Kobe, Japan, earthquake. The properties of the primary and secondary synthetic pulses are listed in Table 3. The three synthetic pulses utilized to fit the KOB record are in fairly good agreement with the individual contributions of the three fault segments as identified by Wald (1996). (Parenthetically, we should point out that we could not get satisfactory fitting with a single pulse, even though different combinations of input parameters were tested. On the other hand, we should state that the fitting that we achieved with more than one pulse may not be unique.)

Cursory inspection of the parameter values listed in Table 3 reveals that γ varies from a value slightly larger than 1 up to a maximum value of 3. Whenever permanent dis-

placement offsets exist in recorded ground motions, γ approaches unity and ν gets values close to 0° or 180° (i.e., symmetric velocity pulse). However, for cases characterized by a number of oscillations, larger γ values are being used. In addition, it should be noted that the pulse period defined by equation (4) and the period of the recorded ground motion at which the pseudovelocity response spectrum exhibits its peak value are fairly close for the majority of the cases considered in this study. Finally, the values of amplitude, A (Table 3), are on average slightly smaller than the corresponding peak ground velocity (PGV) values (Table 1). This is easily explained by the narrowbandedness of the analytical model.

Regression Analysis and Scaling Laws

In order for seismologists and earthquake engineers to generate reliable synthetic near-fault time histories suitable for engineering design using the simple analytical model presented in this article, it is necessary to establish scaling laws for the parameters of the model (i.e., how the parameters scale with earthquake size). Such scaling laws can be derived mainly through regression analysis of existing data and gradually be updated as additional near-source ground-motion records become available.

The model parameters T_p and A are directly related, respectively, to the rise time and slip velocity of faulting, and thus they are expected to scale accordingly. On the other hand, the values that the parameter γ takes do not reveal any systematic relation with earthquake size or closest distance to the fault. This lack of pattern may only partially be explained by the fact that the value of γ depends on the processing of the original “raw” recorded data. For instance, the value of γ approaches unity for time histories recorded by broadband digital strong motion instruments and processed so that their long-period frequency content is preserved (see, for example, records with permanent displacement offsets in Fig. 5). If the same records had been high-pass filtered, thus removing the low-frequency content (i.e., a standard processing procedure for strong motion data recorded by analog instruments), larger γ values would have been necessary to fit the proposed mathematical model to the high-pass filtered ground-motion records. We anticipate that as the number of near-fault recordings increases with time, this problem may be addressed more thoroughly.

The prevailing frequency, f_p (or equivalently, the period, T_p), of the recorded velocity pulses estimated through the aforementioned calibration procedure is strongly correlated with the moment magnitude (M_w). Figure 7a illustrates this variation by denoting the data points differently depending on the type of the causative fault. It is evident that for a given moment magnitude, the duration of pulses produced by strike-slip faults is on average larger than the duration of pulses generated by reverse faults; data points corresponding to oblique faults lie somewhere in between the strike-slip and reverse-fault cases. Figure 7b displays the variation of pulse period with moment magnitude of the

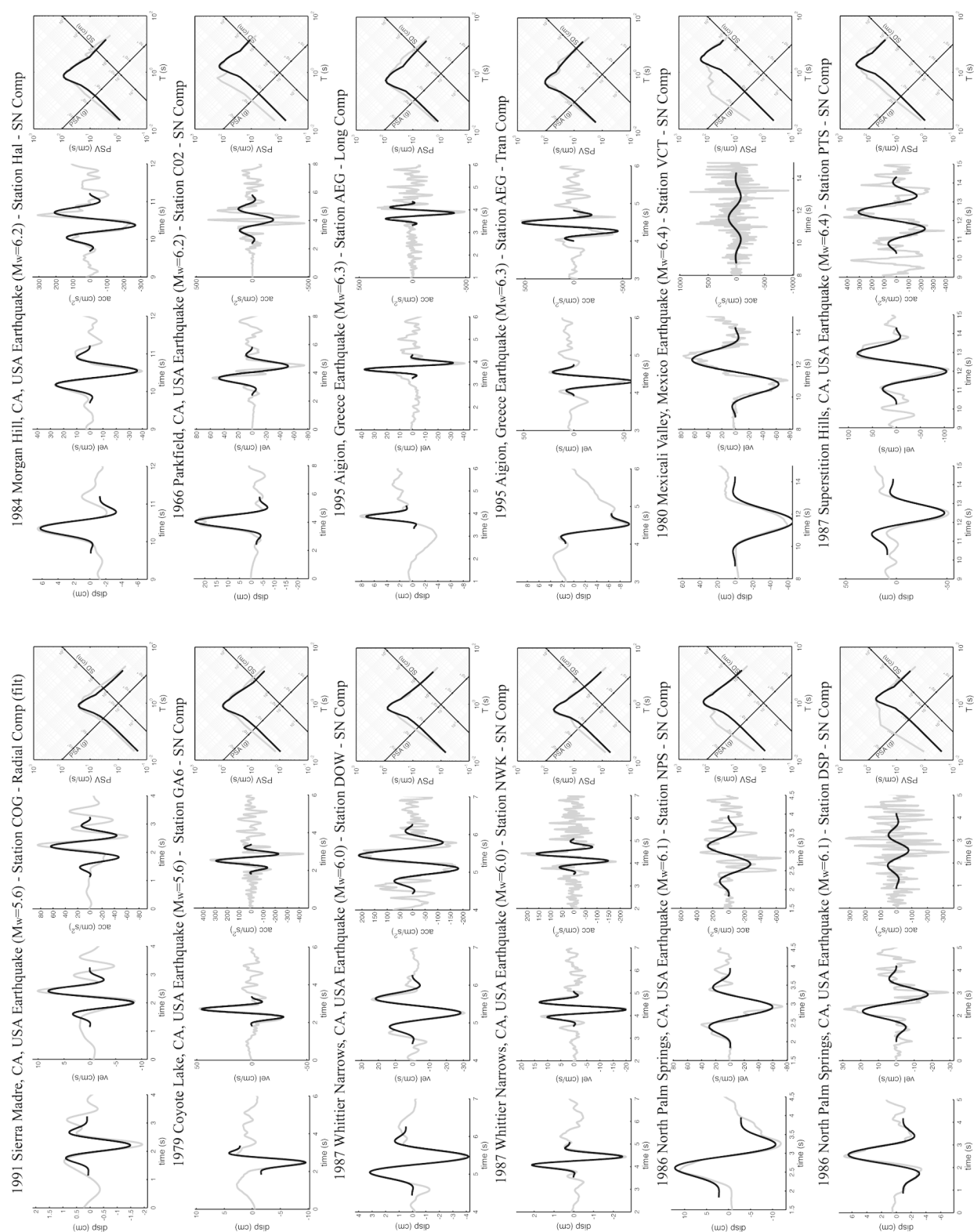


Figure 5. Proposed analytical model (black trace) fitted to recorded ground motions (gray trace). Both time histories (i.e., displacement, velocity, and acceleration series) as well as the corresponding 5% damping elastic response spectra are illustrated. Records are presented in increasing moment magnitude order. (Continued on the following two pages.)

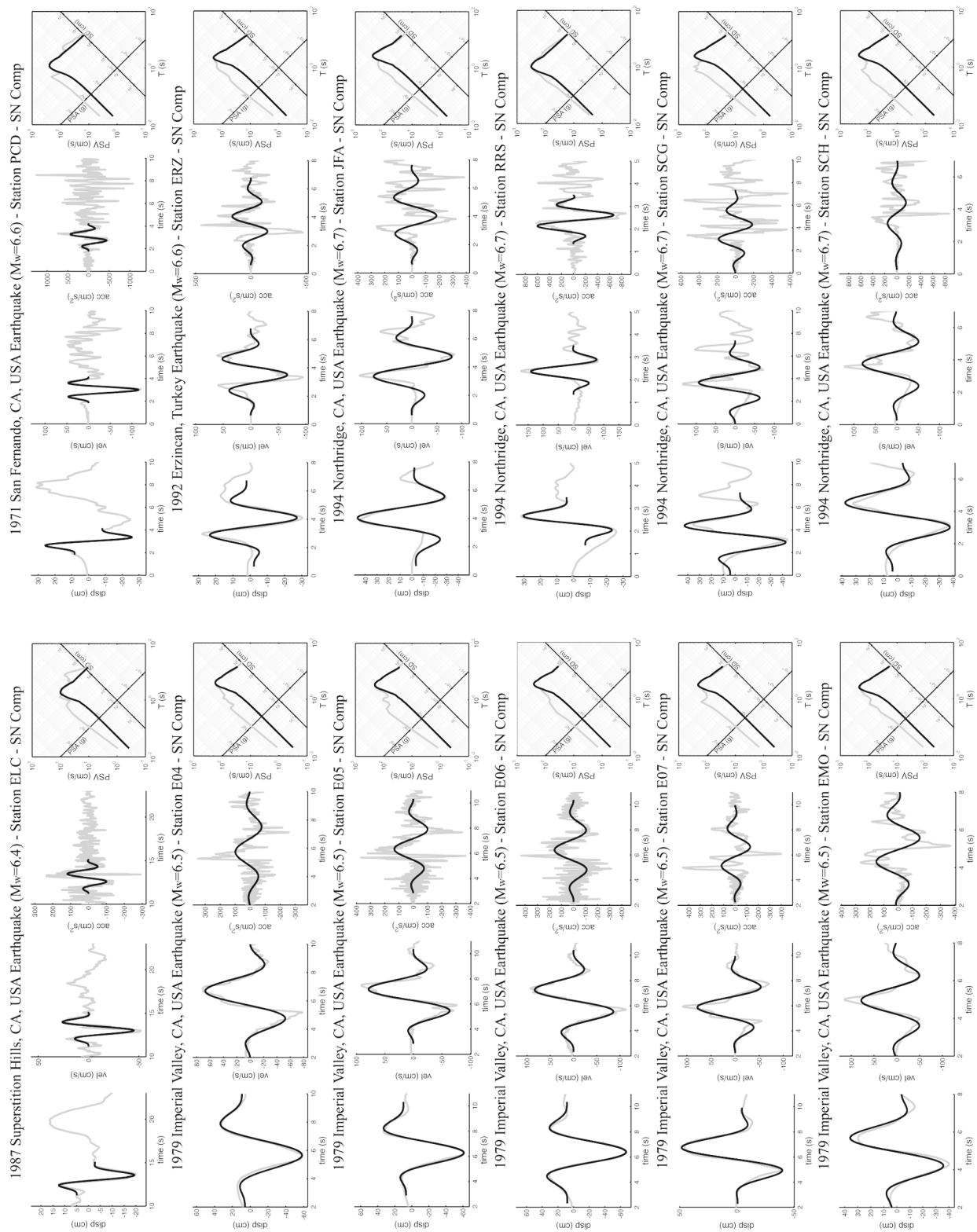


Figure 5. (Continued)

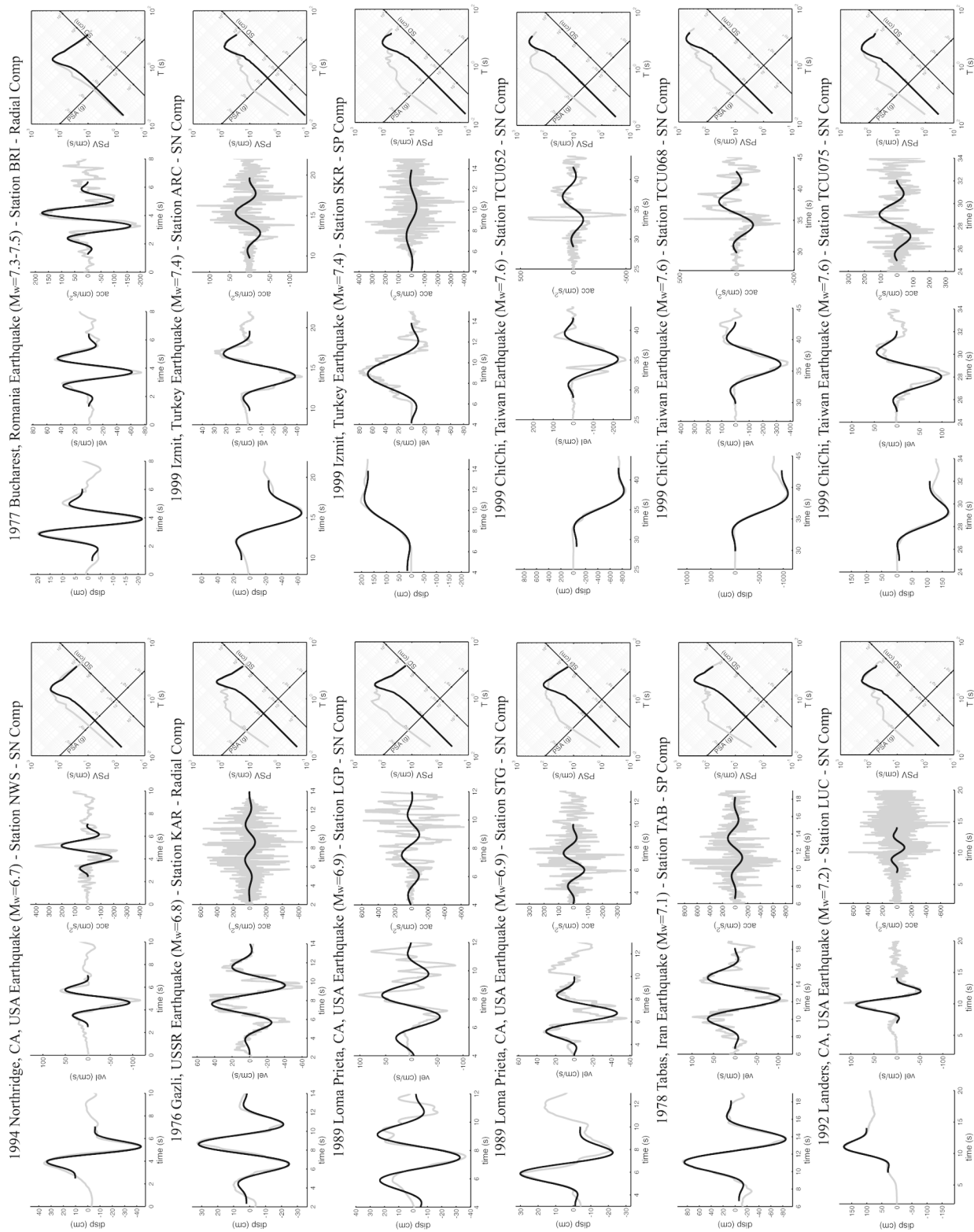


Figure 5. (Continued)

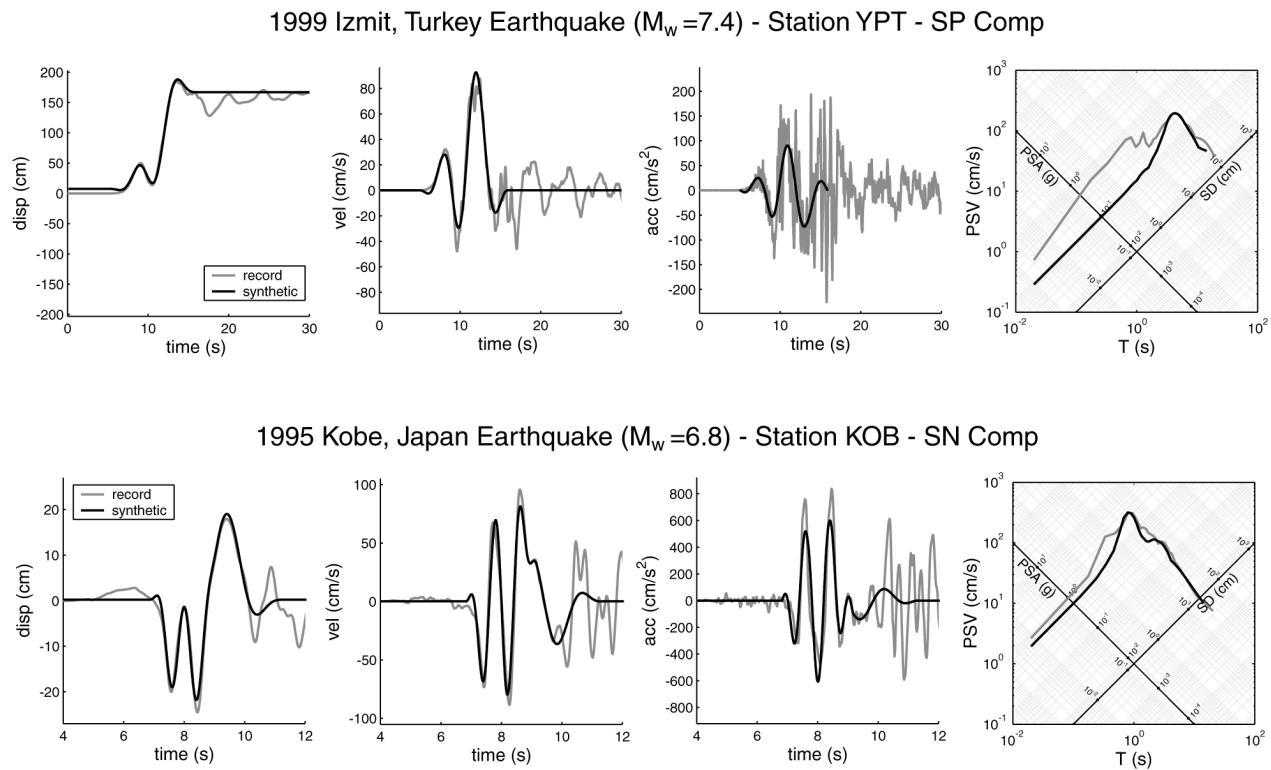


Figure 6. Proposed analytical model fitted to recorded ground motions obtained at (top) YPT station during the 1999 Izmit, Turkey, earthquake (two synthetic pulses have been combined to generate the illustrated synthetic time series) and (bottom) KOB station during the 1995 Kobe, Japan, earthquake (three synthetic pulses have been combined to generate the illustrated synthetic time series).

same group of data; however, in this figure the data points are grouped according to the direction of the recorded ground motion with respect to the strike of the causative fault (i.e., strike-normal, strike-parallel) or with respect to the epicenter (i.e., radial, transverse). The data points that represent the strike-parallel velocity pulses display the same trend as the data points of the near-fault velocity pulses registered in the strike-normal direction.

Figure 7a,b illustrates the complete set of data points of the near-fault records listed in Table 1. However, there are a few exceptions that do not follow the T_p versus M_w variation pattern outlined by the majority of the records. The short-duration velocity pulse (i.e., $T_p = 2.1$ sec) that characterizes the BRI record obtained from the 1977 Bucharest, Romania, earthquake ($M_w 7.3$) is such a characteristic example. This pulse duration is less than half of the typical duration expected for earthquakes of this magnitude. Besides the large depth of the seismic source associated with this particular earthquake, there is an additional feature that distinguishes the BRI record from the rest of the time histories of Table 1: the 1977 Bucharest, Romania, earthquake was an intraplate reverse seismic event. This implies one of the following explanations: (1) relatively larger stress drop, shorter rise time, and therefore shorter duration of the ground velocity pulse for intraplate events as compared to the cor-

responding parameters of interplate events; (2) higher stress drops and stronger rupture directivity (high isochrone velocities) associated with reverse faulting, as compared to strike-slip faulting, (see figure 2 of Aki [1992] and relevant discussion in O'Connell [1998, 1999]); or, more likely, (3) both of these effects.

Other outliers are the pulses recorded during the 1987 Whittier Narrows, California, earthquake ($M_w 5.9$; stations DOW and NWK) and the 1995 Aigion, Greece, earthquake ($M_w 6.3$; station AEG), which are also characterized by shorter periods than it is anticipated for earthquakes of similar magnitudes. The near-fault pulses recorded during the latter two earthquakes may be associated with the *SP* wave (also known as surface or interface *P* wave; see Bouchon, 1978a) generated by stopping phases. Namely, Kawase and Aki (1990) invoked the *SP* wave to explain the damage pattern observed after the 1987 Whittier Narrow, California, earthquake. Mavroeidis and Papageorgiou (2000) explored the possibility that the intense pulses recorded at the horizontal components at the AEG station during the 1995 Aigion, Greece, earthquake may be also associated with the *SP* wave. Furthermore, the waveform of the AEG velocity pulse may have been affected by the strong medium anisotropy that characterizes the Corinth rift (Bouin *et al.*, 1996). Finally, the HAL record obtained from the 1984 Morgan Hill,

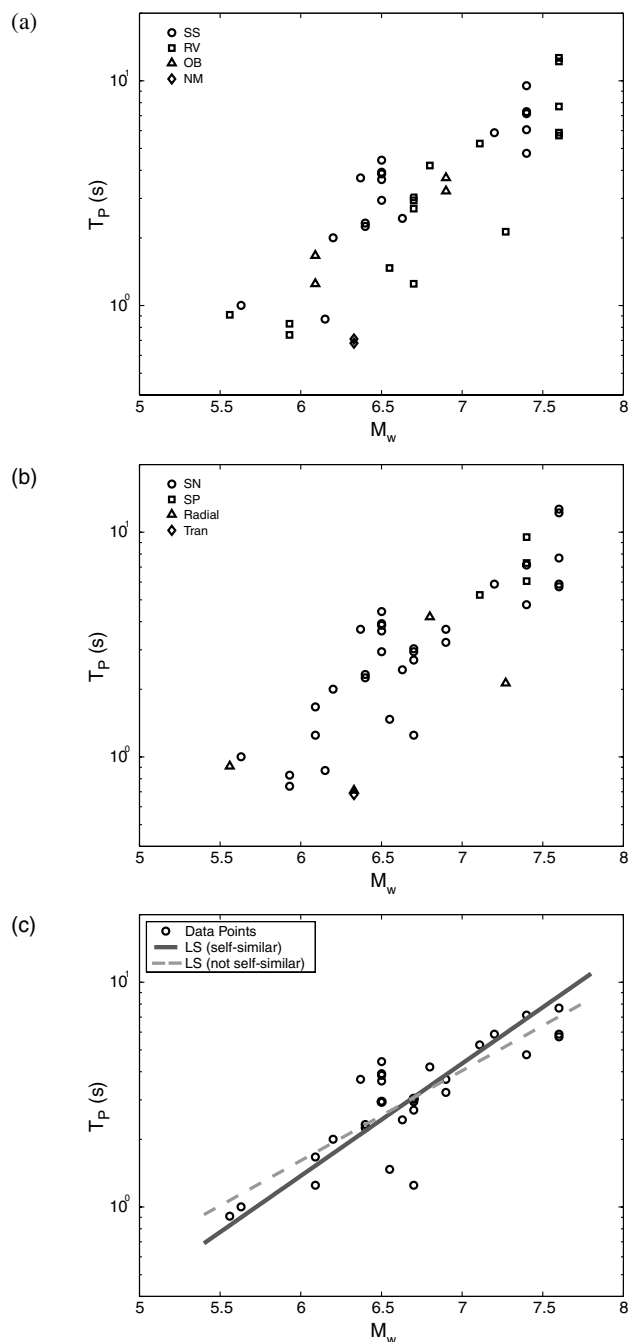


Figure 7. Variation of pulse period (T_P) with moment magnitude (M_w) (a) as function of the fault type (SS, strike-slip; RV, reverse; OB, oblique; NM, normal), (b) as function of the direction of the record (SN, strike-normal; SP, strike-parallel; Rad, Radial; Tran, transverse), and (c) empirical relationships obtained by least-squares fit with and without the assumption of self-similarity based on forward directivity pulses only.

California, earthquake also exhibits a velocity pulse with duration shorter than anticipated for similar magnitude earthquakes. Beroza and Spudich (1988) inferred that rupture was delayed and then progressed back toward the hypocenter during the M_w 6.2 1984 Morgan Hill earthquake; they showed that the HAL record is likely the result of abrupt changes in fault rupture velocity (for an analytical model of the effects of rupture kinematics on high-frequency radiation, see Sato [1994] and Dong and Papageorgiou [2002, 2003]).

Figure 7c displays only those data points of the near-fault records listed in Table 1 that incorporate the forward directivity effect. Records that have been identified to include permanent translation (YPT, SKR, GBZ^(SP) TCU052, TCU068) or are possibly affected by special conditions that give rise to unusually short-duration velocity pulses (BRI, HAL, DOW, NWK, AEG) have been excluded. Assuming that the duration of the pulse is independent of the source–station distance for stations located within ~ 10 km from the causative fault, the pulse period and the moment magnitude are related through the following empirical relationship obtained by least-squares fit analysis:

$$\log T_P = -2.2 + 0.4 M_w. \quad (11a)$$

If we additionally require the empirical relationship to satisfy the self-similarity condition (Aki *et al.*, 1977; Papageorgiou and Aki, 1983b; Aki, 1992; Somerville *et al.*, 1999), the following relation is obtained:

$$\log T_P = -2.9 + 0.5 M_w. \quad (11b)$$

Figure 8 illustrates the variation of the recorded PGV values with respect to the moment magnitude (M_w) and the distance (R) of the site to the causative fault. The complete set of the near-fault records listed in Table 1 has been employed to

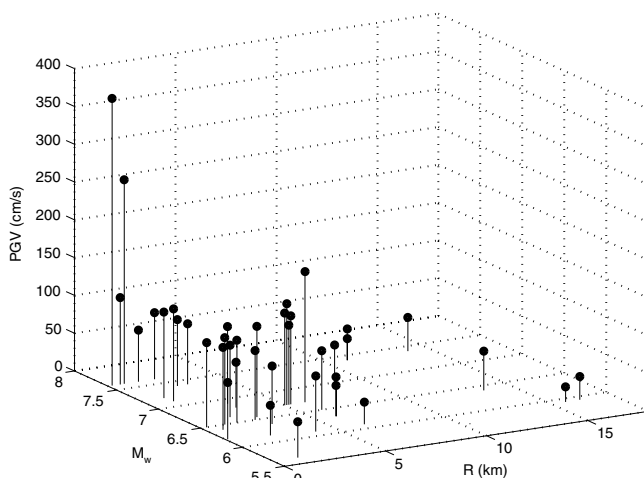


Figure 8. Variation of peak ground velocity (PGV) with moment magnitude (M_w) and distance (R) of station to causative fault.

generate this graph. A similar figure can be obtained with the amplitude, A , of the proposed analytical model substituting for the PGV, as these two quantities are directly related (see previous discussion). The TCU052 and TCU068 records, which are affected by large components of tectonic translation (associated with nearby large-amplitude cumulative fault slip) and possibly by forward directivity as well, are the only velocity pulses included in the compiled database that exhibit unusually intense amplitudes. Figure 8 indicates that the amplitude of the velocity pulses recorded within a distance of 7 km from the causative fault varies from 70 to 130 cm/sec for the entire moment magnitude range considered herein. This observation is in good agreement with the typical slip velocity value of 100 cm/sec frequently considered by seismologists (e.g., Brune, 1970; Aki, 1983). Therefore, a velocity amplitude of 100 cm/sec appears to be a characteristic value that, on average, effectively represents the peak amplitudes of the majority of the available near-fault velocity records. In a future work, we will try to address more thoroughly the dependence of the near-fault PGV on factors such as the station–fault geometry, rupture mode, and so on.

Somerville (1998) derived empirical relationships that relate the pulse period with the moment magnitude. These relationships are very similar to equations (11a) and (11b). Furthermore, Somerville (1998) obtained an empirical relationship for the variation of the PGV as a function of the moment magnitude and the distance of the station to the causative fault, assuming that PGV varies as $1/\sqrt{R}$ for distances larger than 3 km. However, at this point in time, the database is not extensive enough to support with confidence an attenuation relation.

Spectral Properties of the Analytical Model

In this section, the Fourier transform of the proposed analytical model is derived and parametric studies are performed in an effort to represent and physically interpret empirical observations that are based on actual near-fault strong ground motion records. The Fourier transform of the velocity pulse model is given by

$$V(\omega) = \int_{-\infty}^{+\infty} v(t)e^{-i\omega t} dt = R_V(\omega) + i I_V(\omega), \quad (12)$$

where

$$\begin{aligned} R_V(\omega) = & \frac{A\omega_P^2}{2\Phi} \{ \gamma^2 \cos(\pi\gamma) [\sin\theta_3 - \sin\theta_1] \omega^3 \\ & - 3\gamma^2 \omega_P \sin(\pi\gamma) [\cos\theta_3 + \cos\theta_1] \omega^2 \\ & + (3\gamma^2 - 1) \omega_P^2 \cos(\pi\gamma) [\sin\theta_3 - \sin\theta_1] \omega \\ & - (\gamma^2 - 1) \omega_P^3 \sin(\pi\gamma) [\cos\theta_3 + \cos\theta_1] \}, \end{aligned} \quad (13a)$$

$$\begin{aligned} I_V(\omega) = & -\frac{A\omega_P^2}{2\Phi} \{ \gamma^2 \sin(\pi\gamma) [\sin\theta_3 + \sin\theta_1] \omega^3 \\ & + 3\gamma^2 \omega_P \cos(\pi\gamma) [\cos\theta_3 - \cos\theta_1] \omega^2 \\ & + (3\gamma^2 - 1) \omega_P^2 \sin(\pi\gamma) [\sin\theta_3 + \sin\theta_1] \omega \\ & + (\gamma^2 - 1) \omega_P^3 \cos(\pi\gamma) [\cos\theta_3 - \cos\theta_1] \}, \end{aligned} \quad (13b)$$

$$\begin{aligned} \Phi = & (\omega_P - \omega)(\omega_P + \omega) \\ & (-\gamma\omega + \gamma\omega_P + \omega_P)(\gamma\omega + \gamma\omega_P - \omega_P) \\ & (\gamma\omega + \gamma\omega_P + \omega_P)(-\gamma\omega + \gamma\omega_P - \omega_P), \end{aligned} \quad (14)$$

$$\theta_1 = \frac{-\omega\pi\gamma + \nu\omega_P}{\omega_P}, \quad (15a)$$

$$\theta_3 = \frac{\omega\pi\gamma + \nu\omega_P}{\omega_P}, \quad (15b)$$

and

$$\omega_P = 2\pi f_P. \quad (16)$$

A parametric study (in terms of ν and γ) of the normalized (with respect to ω_P and A) Fourier transform of the velocity pulse can be performed based on the equations presented above. We define the normalized frequency as

$$\tilde{\omega} = \frac{\omega}{\omega_P}. \quad (17)$$

Then, the normalized expression of the Fourier transform of the velocity pulse model is obtained by rewriting equations (12) through (16) as

$$\tilde{V}(\tilde{\omega}) = \tilde{R}_V(\tilde{\omega}) + i \tilde{I}_V(\tilde{\omega}), \quad (18)$$

where

$$\begin{aligned} \tilde{R}_V(\tilde{\omega}) = & \frac{R_V(\omega)\omega_P}{A} = \frac{1}{2\Phi} \{ \gamma^2 \cos(\pi\gamma) [\sin\theta_3 - \sin\theta_1] \tilde{\omega}^3 \\ & - 3\gamma^2 \sin(\pi\gamma) [\cos\theta_3 + \cos\theta_1] \tilde{\omega}^2 \\ & + (3\gamma^2 - 1) \cos(\pi\gamma) [\sin\theta_3 - \sin\theta_1] \tilde{\omega} \\ & - (\gamma^2 - 1) \sin(\pi\gamma) [\cos\theta_3 + \cos\theta_1] \}, \end{aligned} \quad (19a)$$

$$\begin{aligned}\tilde{I}_V(\tilde{\omega}) &= \frac{I_V(\omega)\omega_P}{A} = -\frac{1}{2\tilde{\Phi}} \{ \gamma^2 \sin(\pi\gamma)[\sin\theta_3 + \sin\theta_1]\tilde{\omega}^3 \\ &+ 3\gamma^2 \cos(\pi\gamma)[\cos\theta_3 - \cos\theta_1]\tilde{\omega}^2 \\ &+ (3\gamma^2 - 1)\sin(\pi\gamma)[\sin\theta_3 + \sin\theta_1]\tilde{\omega} \\ &+ (\gamma^2 - 1)\cos(\pi\gamma)[\cos\theta_3 - \cos\theta_1] \}, \quad (19b)\end{aligned}$$

$$\begin{aligned}\tilde{\Phi} &= (1 - \tilde{\omega})(1 + \tilde{\omega})(-\gamma\tilde{\omega} + \gamma + 1)(\gamma\tilde{\omega} \\ &+ \gamma - 1)(\gamma\tilde{\omega} + \gamma + 1)(-\gamma\tilde{\omega} + \gamma - 1), \quad (20)\end{aligned}$$

$$\theta_1 = -\tilde{\omega}\pi\gamma + \nu, \quad (21a)$$

and

$$\theta_3 = \tilde{\omega}\pi\gamma + \nu. \quad (21b)$$

Figure 9 illustrates the variation of the normalized Fourier amplitude as function of the normalized frequency for characteristic values of γ and ν . From the representation of the proposed analytical model in the frequency domain, it becomes evident that our mathematical expression produces a narrowband velocity pulse with its bandwidth and Fourier amplitude being a function of γ for constant ω_P and A values. On the other hand, the phase parameter, ν , affects the Fourier amplitude spectrum only when γ approaches unity and then only for periods longer than T_P .

It has been observed (originally by Somerville [2000] and subsequently confirmed by Mavroeidis and Papageorgiou [2002]) that the near-source acceleration response spectra of moderate-to-large earthquakes are stronger than those of large earthquakes in the high-frequency range. This trend is reversed at longer periods. Furthermore, it has been empirically observed that the peak spectral acceleration values of moderate-to-large earthquakes are regularly larger than the corresponding values of very strong earthquakes. The simple analytical model that we propose in the present work can be used to explain the foregoing empirical observation. For this, we make use of a result that Hudson (1962) derived that relates the zero damping pseudospectral velocity $S_{PV}(\xi = 0, \omega)$ of an accelerogram with the amplitude, $|F_{acc}(\omega)|$, of its Fourier transform. Specifically, Hudson demonstrated that $|F_{acc}(\omega)| \leq S_{PV}(\xi = 0, \omega)$. From this it follows easily that $\omega|F_{acc}(\omega)| \leq \omega S_{PV}(\xi = 0, \omega) \Leftrightarrow \omega|F_{acc}(\omega)| \leq S_A(\xi = 0, \omega)$. Hudson (1962) demonstrated that $|F_{acc}(\omega)|$ provides a very good estimate of $S_{PV}(\xi = 0, \omega)$, even though it is a lower bound of it. Thus instead of comparing S_A 's of different-sized earthquakes, we can compare the corresponding $\omega|F_{acc}(\omega)|$. (We are aware that our demonstration is not a proof in the strict mathematical sense. Nevertheless it is an indication that the proposed analytical model is consistent with the above empirical observation. In Mavroeidis *et al.* [2003] we derived closed-form expressions of the response of the elastic oscillator, and thus we worked directly with S_A

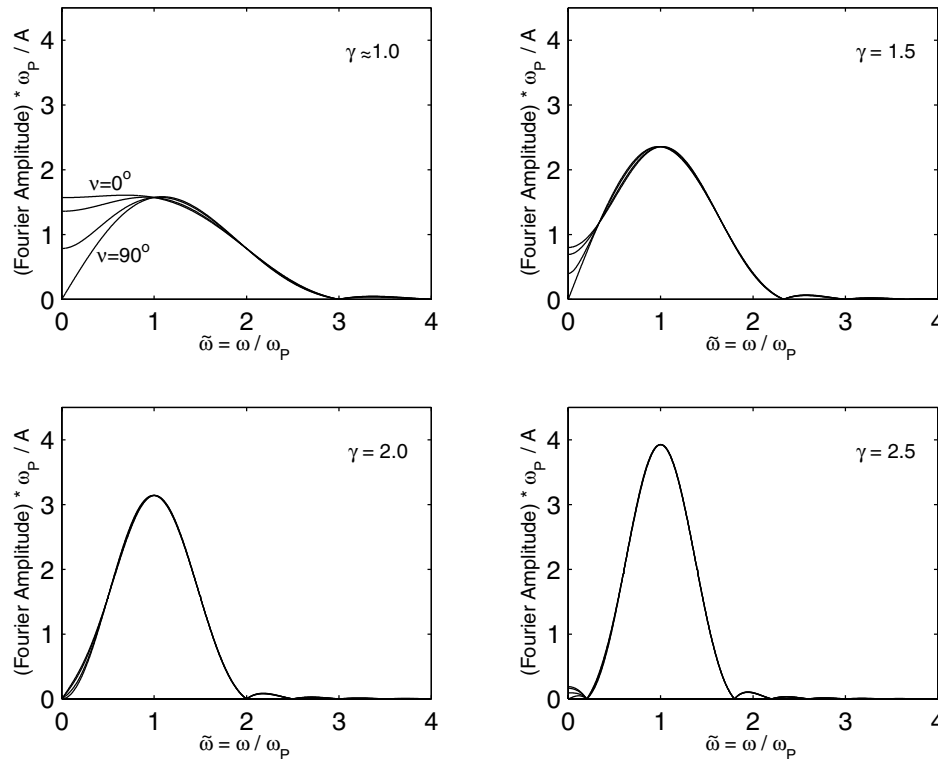


Figure 9. Variation of the normalized Fourier amplitude of the proposed mathematical expression for near-fault velocity pulses as function of the normalized frequency for characteristic values of γ and ν .

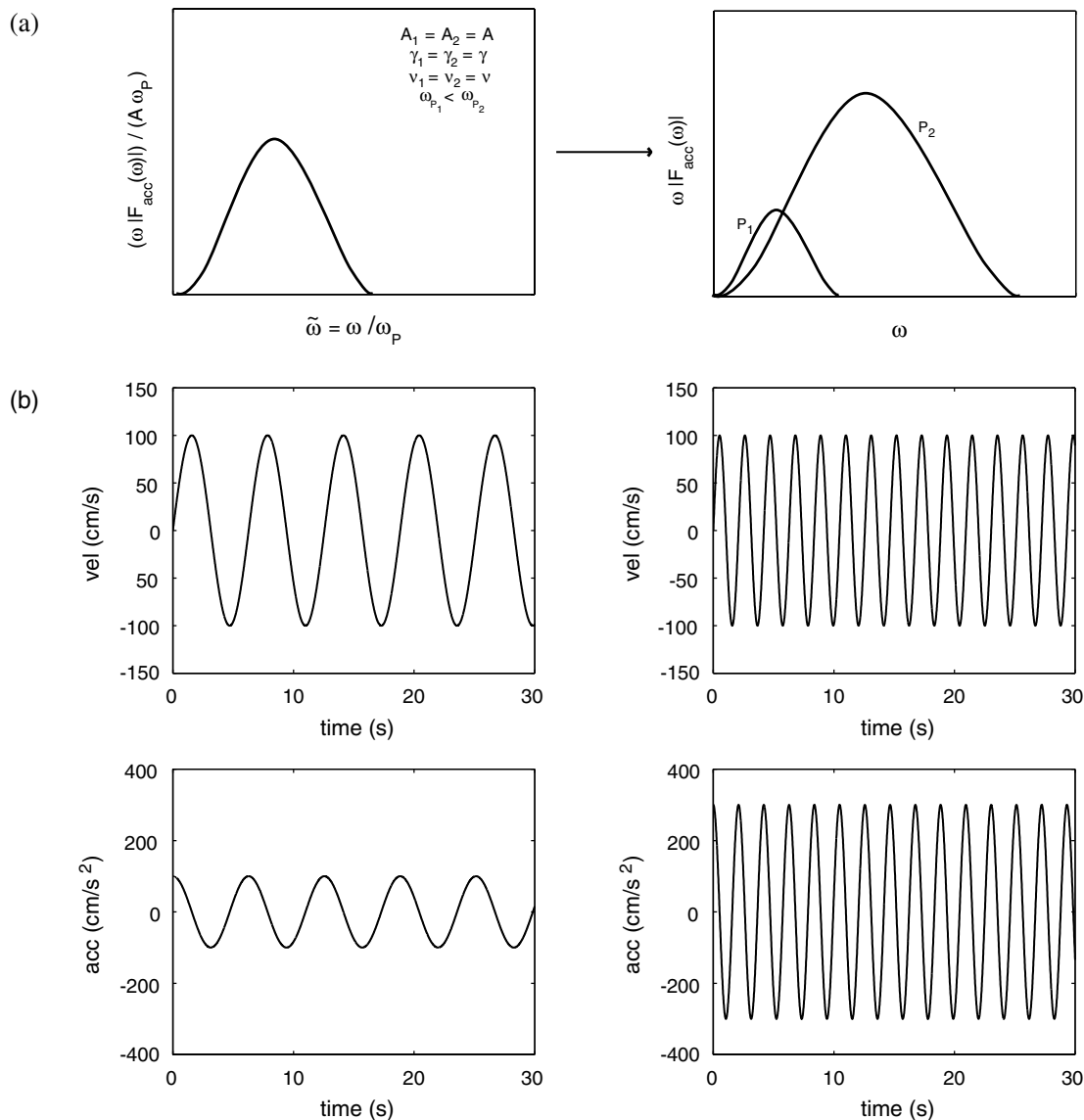


Figure 10. (a) Two hypothetical seismic events with moment magnitudes $M_{w1} > M_{w2}$ that generate near-fault velocity pulses P_1 and P_2 , respectively (we assume $M_{w1} > M_{w2}$ [and thus $\omega_{p1} < \omega_{p2}$], $\gamma_1 = \gamma_2$, $\nu_1 = \nu_2$, and $A_1 = A_2$) are considered to explain that (1) moderate-to-large earthquakes have more energy than very strong earthquakes in the high-frequency range, and (2) peak spectral acceleration values of moderate-to-large earthquakes are regularly larger than the corresponding values of very strong earthquakes. (b) A simplified explanation of panel (a): the amplitude of the derivative of a harmonic signal is proportional to the frequency of the signal.

and presented a more straightforward demonstration of the correspondence of the analytical model to the abovementioned empirical observations.)

In Figure 10a we are considering two hypothetical seismic events with moment magnitudes M_{w1} and M_{w2} that produce near-source ground velocity pulses P_1 and P_2 that are effectively replicated by the proposed analytical model. Let us assume that $M_{w1} > M_{w2}$ (and thus, based on equation 11, $T_{p1} > T_{p2}$ or $\omega_{p1} < \omega_{p2}$), while $\gamma_1 = \gamma_2$, $\nu_1 = \nu_2$, and $A_1 = A_2$. (The assumptions regarding the γ , ν , and A parameters are made for the sake of simplicity and do not influence the conclusion drawn below for the following reason: we

pointed out previously that the amplitude, A , is a fairly stable parameter while γ was found to vary within a limited range ($1.0 < \gamma < 3.0$), and its variation affects the spectral amplitudes by a factor of 2 [Table 3 and Fig. 9]. On the other hand T_p was found to vary from ~ 1 to ~ 8 sec [i.e., factor of 8; see Fig. 7 and Table 3]. Therefore, it is reasonable to conclude that while the variability of the parameters A and γ contributes to the overall variability of the spectral amplitudes, on average the controlling factor is the pulse duration, T_p , or equivalently the prevailing frequency, f_p .) The synthetic signals corresponding to the two earthquakes are represented by identical curves in normalized $\omega |F_{acc}(\omega)|$ versus

normalized frequency $\tilde{\omega}$ plots shown on the left-hand side of Figure 10a. The same spectral amplitudes plotted now on absolute scales are shown on the right-hand side of Figure 10a. It is evident that the larger earthquake exhibits rich energy content in the long-period range, while the smaller earthquake is characterized by high-energy concentration in the high-frequency range.

Figure 10b provides a simplified explanation of Figure 10a. Consider two harmonic signals (representing the near-fault velocity pulses) having the same amplitude. Then, their derivatives (representing the corresponding accelerations of the abovementioned near-fault pulses) have amplitudes proportional to the respective frequencies of the signals. It is apparent that the acceleration time history of the high-frequency velocity signal (i.e., the record of the smaller earthquake) exhibits larger amplitudes than the acceleration time history of the low-frequency velocity signal (i.e., the record of the larger earthquake) for the very simple reason that the amplitude of the time derivative of a harmonic signal is proportional to the frequency of the signal. Consequently, the peak amplitude of the acceleration response spectrum of the short-period time history would be greater than the peak amplitude of the acceleration response spectrum of the long-period time history. Furthermore, the high-energy content for each response spectrum will show up in the vicinity of the signal period; that is, in the long- and short-period ranges for the low- and high-frequency signals, respectively.

Synthesis of Near-Fault Ground Motions

In this section we propose a very simplified methodology for generating realistic synthetic ground motions that are adequate (in the sense articulated by Housner [1970]; however, see also caveats discussed in the Conclusions) for engineering analysis and design. We exploit the simple analytical model introduced in the present work to describe the coherent (long-period) component of motion and the stochastic (or engineering) approach to synthesize the incoherent (high-frequency) seismic radiation (for a review of the stochastic approach of ground-motion synthesis, see Shinazuka [1988]; see also Boore [1983]). For the latter component of motion, due to the proximity of the point of observation to the source, it is necessary to use a source model that provides guidance as to how to distribute the available seismic moment of the simulated event on the fault plane. Such a source model is the specific barrier model of Papageorgiou and Aki (1983a). According to this model, an earthquake is visualized as a sequence of equal-size subevents uniformly distributed on a rectangular fault plane. The subevents are represented by circular cracks (ω^2 model) that radiate seismic energy as the rupture front sweeps the fault plane. (In passing, we note that stochastic models such as that of Zeng et al. [1994], and various other ω^2 -subevent summation approaches, could be used as well to generate the stochastic portion of the synthetic ground motions.)

At the present time, the proposed mathematical model along with its scaling laws can take into account (with con-

fidence) only the forward directivity effect. Even though the analytical expression can replicate near-fault ground-motion records that manifest the permanent-translation effect as well (e.g., the TCU052, TCU068, and SKR records in Fig. 5 and the YPT record in Fig. 6), the limited number of recordings with permanent translation does not permit the derivation of appropriate scaling laws. Therefore, the proposed analytical model should be utilized with caution for the generation of synthetic long-period ground motions that intend to incorporate the permanent translation effect. In these cases, the permanent offsets of the synthetic displacement time histories should be compatible with the tectonic environment and earthquake magnitude of the simulated event (e.g., Wells and Coppersmith, 1994).

The proposed methodology consists of the following steps:

1. Select the moment magnitude M_w of the potential earthquake and calculate the prevailing frequency, f_p , using equations (11) and (4). For selected values of the parameters A , γ , and ν (or for a suite of values of these three parameters), generate the coherent component of acceleration time history (or a suite of time histories) using equation (5).
2. For the selected fault–station geometry, generate synthetic acceleration time histories for the moment magnitude, M_w , specified previously, using the specific barrier model.
3. Calculate the Fourier transform of the synthetic acceleration time histories generated in steps 1 and 2.
4. Subtract the Fourier amplitude spectrum of the synthetic time history generated in step 1 from the Fourier amplitude spectrum of the synthetic time history produced in step 2.
5. Construct a synthetic acceleration time history so that (1) its Fourier amplitude spectrum is the difference of the Fourier amplitude spectra calculated in step 4 and (2) its phase coincides with the phase of the Fourier transform of the synthetic time history generated in step 2.
6. Superimpose the time histories generated in steps 1 and 5. The near-source pulse is shifted in time so that the peak of its envelope coincides with the time that the rupture front passes in front of the station (Anderson and Luco, 1983).

As a demonstration case we generate ground motion for an M_w 6.8 earthquake event and for the fault–station geometry displayed in Figure 11. The site characterization at the station is assumed to be NEHRP site class C. The selected values for the parameters of the specific barrier model are consistent with those of Chin and Aki (1991) (see also figure 2 of Aki [1992]) inferred for a comparable-size Californian event. Figure 12 displays the synthetic motions (strike-normal component). The top panels display the incoherent component of motion, the middle panels display the coherent component of motion (i.e., the near-source velocity pulse), and the bottom panels show the superposition

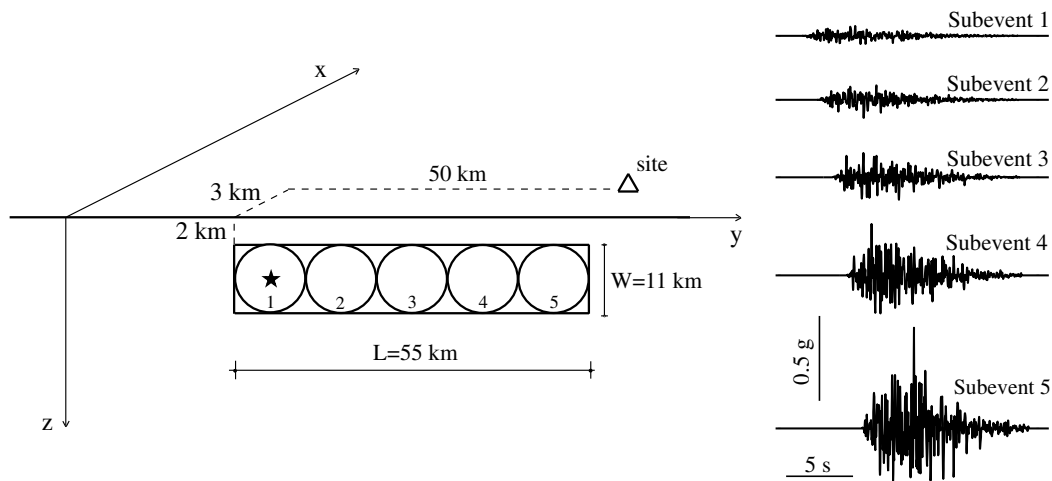


Figure 11. Stochastic simulation of high-frequency ground motion applied to the specific barrier model for a generic M_w 6.8 Californian earthquake with the fault–station geometry illustrated above.

of the above two components. As expected, the incoherent component of motion controls the peak accelerations while the coherent component of motion controls the velocity and displacement.

Conclusions

We have proposed a simple analytical model for the representation of near-field strong ground motions. The model adequately describes the nature of the impulsive near-fault ground motions both qualitatively and quantitatively. In addition, it may be used to analytically explain empirical observations that are based on available near-source records. The input parameters of the model have an unambiguous physical meaning. The proposed analytical model has been calibrated using all the currently available near-field ground-motion records. It successfully simulates the entire set of available near-fault displacement, velocity, and (in many cases) acceleration time histories, as well as the corresponding deformation, velocity, and acceleration response spectra.

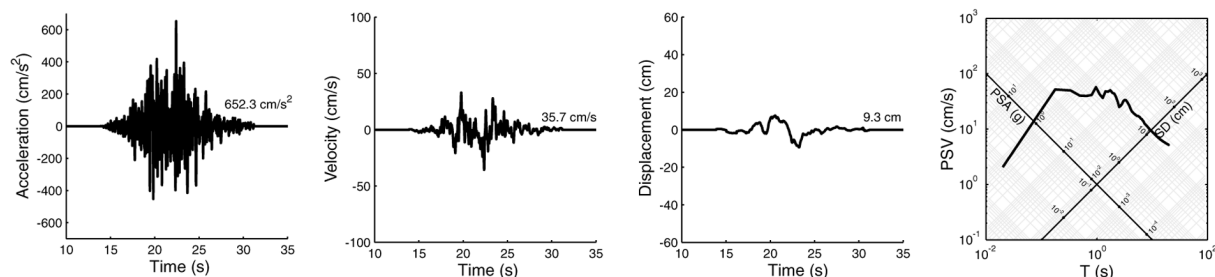
Furthermore, we propose a very simplified methodology for generating realistic synthetic ground motions that are adequate for engineering analysis and design. It should be noted that the analytical model (along with the scaling laws of its parameters) proposed in the present work will facilitate the study of the elastic and inelastic response of conventional and nonconventional (i.e., base-isolated) structures subjected to near-source seismic excitations, as a function of the model input parameters.

From the point of view of an engineer/designer, the proposed model has advantages over simulation methods that are based on either kinematic or dynamic faulting models because the key parameters that control structural response (i.e., T_p and A) appear explicitly in the model equations. Thus, it is straightforward for a designer to examine the sensitivity of the structural response to variations of these

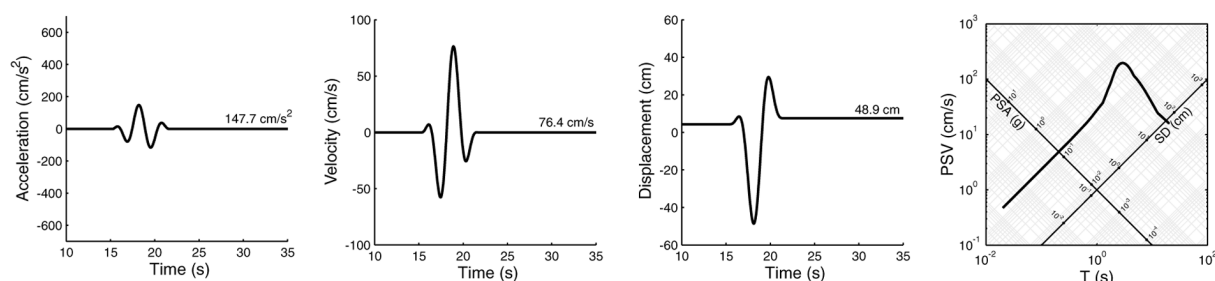
parameters from their mean values for given earthquake magnitude. (Parenthetically we mention that the above advantages are not lost when we hybridize the coherent component of motion with the stochastic/incoherent one in the proposed simplified ground-motion synthesis method for the following reason: structures that are strongly affected by near-fault motions have characteristic period(s) near T_p . It should be evident from Fig. 12 that structural response in the vicinity of T_p is virtually exclusively controlled by the coherent component of motion [i.e., near-fault pulse].) On the contrary, the characteristics of the simulated motions that are based on either kinematic or dynamic faulting models are implicitly controlled by the slip and stress distribution on the fault plane of the simulated event. Thus, variation of the characteristics of the simulated motions that strongly affect structural response can only be quantified by generating an ensemble of synthetic motions by properly varying key source parameters (such as dimensions of subevents, stress drops, etc.). Nevertheless, simulation methods based on kinematic and dynamic faulting models are very useful because they help us discern the physics of ground-motion generation, which is something that the phenomenological model proposed in the present work cannot do.

However, certain caveats must be taken into consideration. The proposed mathematical representation of the near-fault pulses is a purely phenomenological model, and as such it is only as good as the database that was used for its calibration. Factors such as the 3D heterogeneity of the lithosphere, 3D basin responses, source complexity, and so on, may have a substantial influence on the near-fault pulses (e.g., O'Connell, 1998, 1999) (for a recent review and discussion of the subject, see also O'Connell and Ake [2002]). To the extent that variations of factors such as the above are not properly reflected in the existing database (which, after all, is limited at present), then future earthquake events (and their near-fault ground motions) may still hold surprises.

Synthetic Ground Motions (High-Frequency Components)



Synthetic Ground Motions (Long-Period Components)



Synthetic Ground Motions (Combined)

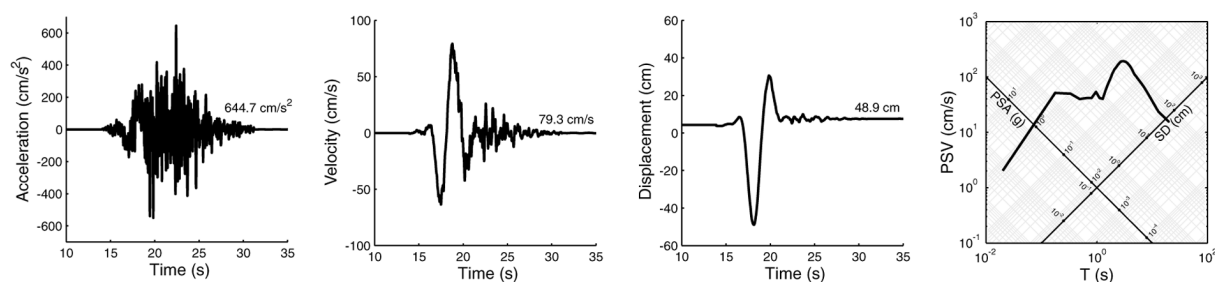


Figure 12. Synthesis of near-fault ground motions for the generic earthquake illustrated in Figure 11.

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Appendix

We have compiled a near-fault strong ground motion database in an attempt to gather and systematically study a large number of near-source ground-motion time histories. Informative tables that summarize database parameters pertaining to the source mechanisms and the ground motion records is presented. We have also included, to the best of our knowledge, a comprehensive list of references related to the seismic events of the database with emphasis on publications associated with inversions of slip distributions on the causative fault planes.

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Table A1
Near-Fault Strong Ground Motion Database

Location	Date	Mechanism	Moment Mo (x 10 ²⁵ dyn-cm)	Mw	Length (km)	Width (km)	Depth of Top (km)	Depth of Bottom (km)	Dip (°)	Strike (°)	Time Window of Slip (sec)	Rupture Velocity (km/sec)	Slip Model Reference †	Data used for Source Modeling
1 San Francisco, CA, USA	18-Apr-06	SS	400-750	7.7-7.9	300-490	12	0	12	90	N15W, N35W, N50W	1.4 to 4.0	2.7	Wald et al., 1993	Teleseismic, Calibrated Recent Earthquakes
2 Kanto, Japan	1-Sep-23	SS	750	7.9	470	10	0	10	90	variable	-	-	Thacher et al., 1997	Geodetic
3 Imperial Valley, CA, USA	19-May-40	RV	700-800	7.8-7.9	130	70	2	31.6	25 NE	N70W	4.0 to 8.0	3.0	Wald & Somerville, 1995	Geodetic, Teleseismic
		SS	2.9-4.4	6.2-6.4	25	7	0	7	90	N323E	-	infinite	Trifunac, 1972	Strong Motion
			11-14	6.6-6.7									Trifunac & Brune, 1970	
4 Olympia, WA, USA	13-Apr-49	SS	11-15	6.6-6.7	0	0	54	54	45 N	N90E	12.5	3.5	Baker & Langston, 1987	Teleseismic, Regional
5 Puget Sound, WA, USA	29-Apr-65	NM	14	6.7	0	0	60	60	70 NE	N344E	3.0	-	Langston, 1981	Strong Motion, Teleseismic
6 Parkfield, CA, USA	27-Jun-66	SS	1.4	6.0	30-37	3	0	3	90	N140E	- 0.3	2.2	Aki, 1968	Strong Motion
		SS	4.4	6.4	20	6	3	9	85.5 SW	N137E	0.7	2.4-2.5	Trifunac & Udvardi, 1974	Strong Motion, Geodetic
		SS	2.3	6.2	30	6	3	9	90	N140E	-	3.1	Bouchon, 1979	Strong Motion
		SS	-	-	infinite	2	0.1	2.1	90	N141E	0	2.2	Archuleta & Day, 1980	Strong Motion
		SS	4.4	6.4	42	12	0	12	90	N140E	-	-	Mendez & Luco, 1990	Strong Motion
7 Koyna, India	10-Dec-67	OB	3.2	6.3	-	-	-	-	67 E	N16E	6.5	-	Segall & Du, 1993	Geodetic
													Langston, 1976	Teleseismic
													Gupta et al., 1969	
													Mandal et al., 1998	
8 San Fernando, CA, USA	09-Feb-71	RV	15.3	6.7	10	10	5	13	40 NE	N288E	0.7	2.0	Trifunac, 1974	Strong Motion
		RV	8.6	6.6	10	10.5	0	5	53 NE	N290E	0	1.8	Langston, 1978	Teleseismic
		RV	7	6.5	18.67	16	3	15.9	29 NE	N280E	0	1.8		
		RV	10	6.6	17.7	11.3	0	8	54 NE	N290E	0.8	2.8	Heaton, 1982	Strong Motion, Teleseismic, Geodetic
9 Gazli, USSR	17-May-76	RV	16	6.7	10	18.5	0	15	54 SE	N40E	8.0	2.0 to 2.5	Hartzell, 1980	Teleseismic
		RV	24	6.9	-	-	-	-	56 SE	N25E	7.0	-	Kristy et al., 1980	Teleseismic
10 Bucharest, Romania	04-Mar-77	RV	100-200	7.3-7.5	0	0	100	100	70 to	N40E	< 1.0	3.8	Hartzell, 1979	Strong Motion, Teleseismic
11 Tabas, Iran	16-Sep-78	RV	58	7.1	90	45	1	20	25 NE	N330E	0.7 to 2.1	2.5	Hartzell & Mendoza, 1991	Strong Motion, Teleseismic
													S.-Taheri & Anderson, 1988	
													Salik, 1994	
12 Coyote Lake, CA, USA	8-Jun-79	SS	0.35	5.6	10	10	2	11.85	80 NE	N336E	1.0	2.8	Liu & Helmberger, 1983	Strong Motion
13 Imperial Valley, CA, USA	15-Oct-79	SS	5	6.4	36	10.5	0	10.5	90	N143E	1.0	2.6	Hartzell & Helmberger, 1982	Strong Motion
		SS	9.1	6.6	50	10	0	10	90	N143E	0.75-3.75	variable	Olson & Apsel, 1982	Strong Motion
		SS	5	6.4	42	10.5	0	10.5	90	N143E	0.7	2.5	Hartzell & Heaton, 1983	Strong Motion, Teleseismic
		SS	6.7	6.5	35	13	0	12.8	80 NE	N143E	0 to 1.9	variable	Archuleta, 1984	Strong Motion
		SS	7	6.5	40	9.8	0.4	10	80 NE	-	0	3.0	Mendez & Luco, 1990	Strong Motion
													Bouchon, 1997	
14 Mexicali Valley, Mexico	9-Jun-80	SS	4.5	6.4	30	9.75	0.25	10	90	N137E	0	2.8	Mendez & Luco, 1990	Strong Motion
15 Borah Peak, ID, USA	28-Oct-83	NM	21	6.8	19	22.6	0	16	45 SW	N138E	11	3.0	Doser & Smith, 1985	Teleseismic, Regional Ground Motion
		NM	27	6.9	50	26	0	19.6	49 SW	N152E	-	-	Ward & Barrientos, 1986	Geodetic
		NM	-	-	75	10	-	-	45 SW	N110E-	-	-	Richins et al., 1987	Teleseismic
		NM	23	6.8	52	26.4	1	20.9	49 SW	N170E	0.6	2.9	Mendoza & Hartzell, 1988	Teleseismic
		NM	33.4	7.0	75	26	0	19.6	49 SW	N152E	-	-	Du et al., 1992	Geodetic
16 Morgan Hill, CA, USA	24-Apr-84	SS	2.1	6.2	27	11.5	0.5	12	90	N148E	0.3	0.9 β	Hartzell & Heaton, 1986	Strong Motion, Teleseismic
		SS	2.7	6.2	30	10	2.5	12.5	85 NE	N150E	0.2	0.8 β	Beroza & Spudich, 1988	Strong Motion
													Mikumo & Miyatake, 1995	
													Bouchon, 1997	
17 Michoacan, Mexico	19-Sep-85	RV	1400	8.0	180	140	6	40	14 NE	N300E	2.0	2.6	Mendoza & Hartzell, 1988	Teleseismic
		RV	1500	8.1	180	140	6	40	14 NE	N300E	2.0	2.6	Mendoza & Hartzell, 1988	Strong Motion, Teleseismic

Table A1

Near-Fault Strong Ground Motion Database (Continued)

Location	Date	Mechanism	Moment Mo (x 10 ⁻²⁵ dyn-cm)	Mw	Length (km)	Width (km)	Depth of Top (km)	Depth of Bottom (km)	Dip (°)	Strike (°)	Time Window of Slip (sec)	Rupture Velocity (km/sec)	Slip Model Reference †	Data used for Source Modeling
18 Nahanni, Canada	23-Dec-85	RV	15	6.7	48	21.3	2	11	25 SW	N160E	3.0	2.75	Yomogida, 1988 Campillo et al., 1989 Ruppert & Yomogida, 1992	Strong Motion, Telesismic
19 Palm Springs, CA, USA	8-Jul-86	OB	1.7 1.7 to 1.8	6.1 6.1	22 22	15.2 15.2	4 4	15 15	46 NE 46 NE	N287E N287E	0.6 0.4	3.0 3.0	Mikumo et al., 1988 Hartzell et al., 1994 Weichert et al., 1986	Telesismic
20 Whittier Narrows, CA, USA	10-Oct-87	RV	1 1	6.0 6.0	10 12	10 12	12.1 11.2	17.1 18.0	30 NE 35 NE	N280E N270E	0.2-0.4 0.2	2.5 2.5	Mendoza & Hartzell, 1988 Hartzell, 1989 Hartzell & Iida, 1990 Zeng et al., 1993	Strong Motion, Telesismic Strong Motion Strong Motion
21 Elmore Ranch, CA, USA	24-Nov-87	SS	2.3	6.2	25	10	0	10	90	N40E	-	-	Larsen et al., 1992	GPS, Surface Slip
22 Superstition Hills, CA, USA	24-Nov-87	SS	4.8 9.4	6.4 6.6	20 25	11.5 10	0.5 0	12 10	90 90	N127E N130E	0.5 to 1.5 -	2.4 -	Wald et al., 1990 Larsen et al., 1992	Strong Motion GPS, Surface Slip
23 Loma Prieta, USA	17-Oct-89	OB	30 35 23 29 23	6.9 7.0 6.8 6.9 6.8	40 38 40 40 40	20 17 14 14 16	1.5 3 4.8 4 4	20.3 19 18 18.9	70 SW 70 SW 70 SW 70 SW 69 SW	N128E N126E N130E N130E N129E	0.7 to 1.9 1.0 0.3 0.3 1.0 to 6.0 variable	2.7 3.0 0.8* 2.5 variable	Wald et al., 1991 Steidl et al., 1991 Beroza, 1991 Zeng et al., 1993 Horton, 1996 Bouchon, 1997	Strong Motion, Telesismic Strong Motion Strong Motion Strong Motion Strong Motion
24 Sierra Madre, CA, USA	28-Jun-91	RV	0.28	5.6	7	6	9.4	14	50 NW	N242E	0.2 to 0.4	2.7	Wald, 1992	Strong Motion, Telesismic
25 Erzincan, Turkey	13-Mar-92	SS	11	6.6	30	9	3	12	90	N125E	0.35	3.0	Bernard et al., 1997	Strong Motion, Regional Ground Motion
26 Joshua Tree, CA, USA	23-Apr-92	SS	1.7	6.1	35	20	0	20	90	N172E	5.0	2.4-3.2	Bennett et al., 1995	Geodetic
27 Petrolia, CA, USA	25-Apr-92	RV	1.9 27	6.1 6.9	30 32	20 32	0 4.2	20 11.94	90 14 NE	N160E N350E	3.0 to 5.0 0.3-1.4 variable	3.0 2.8	Hough & Dreger, 1995 Oglesby & Archuleta, 1997	Regional Ground Motion Strong Motion
28 Landers, CA, USA	28-Jun-92	SS	70 to 80	7.2	30	15	0	15	90	N355E	1.0 to 6.0	2.7	Wald & Heaton, 1994	Strong Motion, Telesismic, Geodetic
		SS	60 to 80	7.1-7.2	27	15	0	15	90	N334E	1.0 to 6.0	2.7	Cocher & Beroza, 1994	Strong Motion
		SS	90	7.3	30	18	0	18	90	N354E	2.0 to 6.0	2.5	Cotton & Campillo, 1995	Strong Motion
		SS	66	7.1	27	18	0	18	90	N331E	2.0 to 6.0	2.5	Hartzell & Liu, 1996	Telesismic
		SS	66	7.1	27	18	0	18	90	N322E	1.0 to 5.0 variable	2.5-3.0 2.5-3.0 2.5-3.0 variable 2.5-3.5 variable	Olsen et al., 1997 Bouchon et al., 1998 Day et al., 1998	Strong Motion, Telesismic, Geodetic
29 Northridge, CA, USA	17-Jan-94	RV	17.1	6.8	20	26	1	18.1	41 SW	N110E	-	-	Hudnut et al., 1996	GPS - Variable Slip, GPS - Uniform Slip
		RV	13.4	6.7	30	10.7	0	9.1	58 SW	N122E	-	-	Shen et al., 1996	GPS, Geodetic
		RV	13	6.7	18	24	5	20.4	40 SW	N119.3E N122E	0.6 to 1.4	3.0	Wald et al., 1996	Strong Motion, Telesismic, Geodetic, GPS
		RV	15	6.7	20	24.9	5	21.0	40 SW	N122E	0.5 to 1.5 variable	2.0 to 3.0 variable	Hartzell et al., 1996 Bouchon, 1997	Strong Motion

Table A1
Near-Fault Strong Ground Motion Database (Continued)

Location	Date	Mechanism	Moment Mo ($\times 10^{25}$ dyn-cm)	Mw	Length (km)	Width (km)	Depth of Top (km)	Depth of Bottom (km)	Dip ($^{\circ}$)	Strike ($^{\circ}$)	Time Window of Slip (sec)	Rupture Velocity (km/sec)	Slip Model Reference †	Data used for Source Modeling
30 Hanshin (Kobe), Japan	17-Jan-95	SS	22	6.8	24	16	0.005	15.46	75 SE	N43E	1.0 to 3.0	2.5	Day et al., 1998 McGarr & Fletcher, 2000	Geodetic, Strong Motion, Teleseismic
		SS	19.1	6.8	36	16	0.1	16.04	85 NW	N52E	1.0 to 3.0	2.5	Yoshida et al., 1996	Geodetic, Strong Motion, Ground Motion
					21	21	0	21.8	82 SE	N45E	0.8 to 4.3	3.1	Sekiguchi et al., 1996	
					15	21.8	0	21.8	90	N53E	0.8 to 4.3	3.1		
		SS	24	6.9	24	21.9	0	21.8	85 NW	N233E	0.8 to 4.3	3.1	Wald, 1996	Geodetic, Strong Motion, Teleseismic
					20	20	0.2	19.9	80 SE	N45E	0.7 to 2.7	2.8		
					40	20	0.2	20.1	85 NW	N50E	0.7 to 2.7	2.8	Sekiguchi et al., 2000 Pitarka et al., 1998	Geodetic, Strong Motion, Ground Motion
													Day et al., 1998	
31 Aigion, Greece	15-Jun-95	NM	3.4-3.9	6.3	15	9			33 NE	N277E	-	2.7	Bouchon et al., 1998 Bernard et al., 1997	Geodetic, Strong Motion, Regional Ground Motion, Teleseismic
32 Izmit, Turkey	17-Aug-99	SS	170	7.4	168	21	0	21	90	variable	-	-	Reilinger et al., 2000	Geodetic (four satellite systems)
		SS	184	7.5	162	21	0	21	90	variable	-	-	Feigl et al., 2002	Strong Motion
		SS	250	7.5	155	17	0	17	90	variable	1.0 to 4.0	3.0 to 4.8	Bouchon et al., 2002	Strong Motion, Teleseismic
		SS	240	7.5	52.5	22.5	0	22.4	85 N	N84E	1.0 to 7.0	3.5	Delouis et al., 2002	SAR, GPS
					67.5	22.5	0	22.4	85 N	N94E		3.5		
					30	22.5	0	22.4	85 N	N67E		3.5		
					22.5	22.5	0	22.4	85 N	N91E		3.5		
		SS	152	7.4	45	23.3	0	23.3	90	N89E	1.0 to 7.0	3.0	Sekiguchi & Iwata, 2002	Strong Motion
					30	23.3	0	23.3	90	N91E		5.8		
					36	23.3	0	23.3	90	N95E		3.0		
					30	23.3	0	23.3	90	N70E		3.0		
33 Chi-Chi, Taiwan	20-Sep-99	RV	200-450	7.5-7.7	80	40	0	19.4	29 SE	N3E	1.0 to 12.0	2.5	Li et al., 2002	Teleseismic
		RV	290	7.6	25	40	0	19.4	29 SE	N60E	1.0 to 12.0	2.5	Ma et al., 2001	Strong Motion, Teleseismic
		RV	270	7.6	72	44	0	22.25	30 SE	variable	2.0 to 10.0	2.6	Zeng & Chen, 2001	Strong Motion, GPS
					12	43.2	0	22.25	31 SE	N5E	variable	2.5	Wu et al., 2001	Strong Motion, GPS
					12	20	0	10.25	25 SE	N22E				
		RV	410	7.7	105	40	0	20	30 SE	N65E	3.0 to 16.5	2.6	Chi et al., 2001	Strong Motion
		RV	730	7.8	66	40	0	20	30 SE	N5E	-	-	Wang et al., 2001	Surface Displacement (from GPS and Strong Motion)
		RV	540	7.8	70	50	0	25.75	31 SE	N9E	-	-	Yoshioka, 2001	GPS
34 Duzce, Turkey	12-Nov-99	OB	51 to 57	7.1	49	24.5	0	19.8	54 N	N85E	-	-	Burgmann et al., 2002	GPS, SAR

p.s., point source

*When the "Time Window of Slip" parameter is expressed as a time interval, then either multiple time windows have been utilized in the inversion procedure or variable rise time has been employed/calculated; the latter case is denoted by the word "variable".
For slip inversions based on teleseismic data, the "Time Window of Slip" parameter may be associated with the duration of the far-field source time function.

†The references in *italics* do not introduce slip models. They either convey useful information related to the focal mechanism of the particular event or present forward simulation results based on the listed slip models.

Table A2

Near-Fault Strong Ground Motion Database

Location	Date	Near-Fault Station	Closest Distance* (km)	Site Code	Directivity	Original Horizontal Components (°)	Rotation Angle (°)	Peak Acceleration SN** (cm/s ²) SP** (cm/s ²) Vert (cm/s ²)	Peak Velocity SN** (cm/s) SP** (cm/s) Vert (cm/s)	Peak Displacement SN** (cm) SP** (cm) Vert (cm)
1 San Francisco, CA, USA	18-Apr-06	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
2 Kanto, Japan	1-Sep-23	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
3 Imperial Valley, CA, USA	19-May-40	Imperial Valley Irrig. Dist., El Centro (ELC)	8.0	SL	B	180	270	204.1	280.5	201.3
4 Olympia, WA, USA	13-Apr-49	Olympia Highway Test Lab (OLY)	56	?	-	358	86	157.2	269.3	90.6
		Seattle Army Base (SAB)	83	?	-	262	272	54.0	71.9	22.0
5 Puget Sound, WA, USA	29-Apr-65	Olympia Highway Test Lab (OLY)	75	?	-	176	266	142.9	146.1	59.9
		Seattle Federal Office Building (SEF)	63	?	-	148	238	52.6	69.0	32.1
6 Parkfield, CA, USA	27-Jun-66	Cholame Shandon Array 2 (C02)	0.1	SL	F	65	-	466.8	75.1	13.7
		Cholame Shandon Array 5 (C05)	5.2	SL	F	355	85	341.6	362.3	135.5
		Cholame Shandon Array 8 (C08)	9.2	SL	F	320	50	242.0	264.9	113.5
		Tambor Station 2 (TMB)	6.5	?	F	205	295	360.5	234.3	133.5
7 Koyuna, India	10-Dec-67	Koyuna Dam (KOYNA)	?	HR	B	Trans	Long	371.6	480.3	240.0
8 San Fernando, CA, USA	09-Feb-71	Paracoma Dam (PCD)	3.0	HR	F	164	254	1266.0	780.4	696.0
9 Gazli, USSR	17-May-76	Karakay Point (KAR)	3.0	SR	N	0	90	596.7	703.9	1240.0
10 Bucharest, Romania	04-Mar-77	Boucharst Research Institute (BRI)	190	SL	F	270	0	279	165.5	107.0
11 Tabas, Iran	16-Sep-78	Tabas (TAB)	1.2	SL	N	344	74	829.5	831.1	675.4
12 Coyote Lake, CA, USA	8-Jun-79	Gilroy Array 1 (GA1)	9.0	HR	F	230	320	89.4	113.9	58.0
		Gilroy Array 2 (GA2)	7.2	SL	F	50	140	164.0	247.6	162.2
		Gilroy Array 3 (GA3)	5.1	SL	F	50	140	240.2	246.9	135.9
		Gilroy Array 4 (GA4)	3.5	SL	F	270	360	287.9	210.7	409.3
		Gilroy Array 6 (GA6)	1.2	SR	F	230	320	435.6	278.9	146.9
		San Martin Coyote Creek (SMCC)	0.2	?	B	160	250	249.2	140.6	101.2
13 Imperial Valley, CA, USA	15-Oct-79	Aeropuerto Mexicali (AEP)	0.4	SL	F	315	45	353.8	268.8	138.9
		Agrarias (AGR)	0.6	SL	N	273	3	303.2	249.7	818.8
		El Centro Array 3, Pine Union Sch (E03)	11.7	SL	F	140	230	227.4	267.0	124.3
		El Centro Array 4, Anderson Rd (E04)	6.0	SL	F	140	230	349.4	462.1	243.1
		El Centro Array 5, James Rd (E05)	2.7	SL	F	140	230	368.0	518.9	526.9
		El Centro Array 6, Huston Rd (E06)	0.3	SL	F	140	230	433.9	389.0	1624.0
		El Centro Array 7, Imperial Val. Coll. (E07)	1.8	SL	F	140	230	453.1	327.6	533.8
		El Centro Array 8, Cruickshank Rd (E08)	5.0	SL	F	140	230	463.4	575.0	430.9
		El Centro Array 10, Hospital (E10)	9.6	SL	F	320	50	174.3	224.2	103.3
		Bonds Corner (BCR)	11.0	SL	F	140	230	274	745.3	583.0
		Calexico Fire Station (CXO)	2.0	SL	F	225	315	255.7	212.6	183.6
		Mexicali Casa Flores (MXC)	10.4	SL	N	270	0	453.7	303.8	372.1
		Compuertas (CMP)	15.5	SL	B	285	15	121.9	191.2	73.5
		Meloland Route Overpass (EMO)	1.2	SL	F	270	0	370.6	263.9	243.3
		Brawley Airport (BRA)	8.2	SL	F	225	315	164.9	201.9	143.2
		Differential Array -Dogwood Rd (EDA)	6.3	SL	F	270	0	405.8	437.3	693.9
		Post Office, Holtville (HVP)	6.9	SL	F	225	315	253.5	227.9	225.7
		Victoria (VCT)	3.0	SL	F	320	50	929.3	855.5	N/A
14 Mexicali Valley, Mexicc	9-Jun-80	Cerro Prieto (CPR)	6.5	HR	F	315	45	614.1	583.3	298.5
15 Borah Peak, ID, USA	28-Oct-83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
16 Morgan Hill, CA, USA	24-Apr-84	Anderson Dam, Downstream (AND)	2.6	SL	F	250	340	451.0	267.1	203.5
		Coyote Lake Dam, San Martin (CLD)	0.1	SR	F	195	285	803.9	979.8	376.3
		Halls Valley (HAL)	2.0	SL	B	150	240	308.1	148.8	108.0
17 Michoacan, Mexico	19-Sep-85	Caleita de Campos (CAL)	11.2	HR	F	90	180	135.7	115.5	86.8
		La Villita (VIL)	17.5	HR	F	90	180	97.8	111.7	56.7
		La Union (UNI)	20.2	HR	F	90	180	168.8	140.3	126.9
		Zhuatanejo (ZHU)	16.6	HR	F	270	180	123.6	159.2	98.1
18 Nahanni, Canada	23-Dec-85	Iverson, NW Territories (SITE 1)	9.4	HR	N	280	10	954.6	1219.0	322.0
		Slide Mountain (SITE 2)	5.2	HR	N	240	330	513.9	396.1	296.3
19 Palm Springs, CA, USA	8-Jul-86	Cabazon Post Office (CAB)	6.0	SL	F	180	270	226.5	196.1	356.1
		Desert Hot Springs (DSP)	6.4	SL	N	0	90	336.2	280.1	535.4
		North Palm Springs Post Office (NPS)	4.0	SL or SR	F	210	300	656.9	604.0	426.5
		Palm Springs Airport (PSA)	10.8	SL	F	0	90	157.1	151.0	182.0
		Whitewater Trout Farm (WWT)	6.5	SL	F	180	270	524.0	591.5	462.2
20 Whittier Narrows, CA, USA	10-Oct-87	Alhambra, Fremont School (ALF)	13.1	SL	N	180	270	290.7	370.7	173.6
		San Marino, SW Academy (SNM)	15.8	SL	N	270	360	196.1	128.7	101.8
		Los Angeles, Obregon Park (OBP)	12.7	SL	F	270	360	391.3	389.3	125.2
		Alladena, Eaton Canyon Park (ECP)	19.1	SL	N	0	90	309.9	149.1	161.6
		Downey, County Maint. Bldg. (DOW)	16.4	SL	F	180	270	125.2	148.1	149.1
		Garvey Reservoir (GVR)	12.1	SR	F	330	60	291.2	488.6	373.5

Table A2

Near-Fault Strong Ground Motion Database (Continued)

Location	Date	Near-Fault Station	Closest Distance* (km)	Site Code	Directivity	Original Horizontal Components (°)	Rotation Angle (°)	Peak Acceleration		Peak Velocity		Peak Displacement	
								SN** (cm/s ²)	SP** (cm/s ²)	SN** (cm/s)	SP** (cm/s)	SN** (cm)	SP** (cm)
21 Elmore Ranch, CA, USA	24-Nov-87	Whittier Narrows Dam - Upstream (WND)	12.3	SL	F	62	152	246.4	275.7	14.4	14.0	1.7	1.2
		Alhambra 900 S. Fremont (ALH)	14.3	SL	N	0	90	242.2	297.8	19.7	12.8	2.9	1.9
		Whittier 7215 Bright Ave (WTR)	11.1	SL	F	90	180	442.1	573.6	22.6	31.4	3.7	2.9
		Los Angeles - Bulk Mail Center (LBM)	12.7	SL	F	280	10	322.1	436.9	13.9	35.8	1.5	0.4
		Vernon 4814 Loma Vista Ave (VRN)	13.6	SL	F	277	7	273.7	236.9	18.5	20.5	2.6	2.1
		Norwalk 12400 Imperial Highway (NWK)	15.7	SL	F	0	90	224.5	101.6	20.0	12.9	3.5	2.4
		San Gabriel 600 E. Grand Ave (SGS)	14.6	SL	F	0	90	278.4	202.5	22.3	9.7	5.5	3.2
		El Monte 11338 Fairview Ave (EMC)	15.2	SL	N	180	270	126.4	231.4	13.6	5.8	15.4	4.8
		West Covina 1307 S. Orange (XWC)	15.6	SL	N	225	315	121.5	153.8	11.3	7.4	3.7	2.2
		Hacienda Heights 16750 Corna (XHA)	15.0	SL	N	140	230	177.0	216.9	6.7	9.1	2.2	0.8
22 Supersition Hills, CA, USA	24-Nov-87	Aracadia 180 Campus Dr. (ARC)	17.0	SL	N	279	9	291.1	164.6	21.0	6.8	6.2	3.1
		Supersition Mountain (SSM)	9.2	SR or HR	F	45	135	145.4	125.7	111.0	7.9	2.9	1.3
		Parchute Test Site (PTS)	17.1	?	F	225	315	95.6	134.7	5.9	9.1	3.2	2.0
23 Loma Prieta, USA	17-Oct-89	Westmorland Firestation (WST)	15.2	SL	B	90	180	136.2	176.4	9.8	22.5	3.3	4.6
		Parchute Test Site (PTS)	0.7	?	F	225	315	355	350.2	N/A	109.0	46.8	N/A
		Supersition Mountain (SSM)	5.7	SR or HR	N	45	135	710.6	887.4	32.6	41.2	N/A	4.6
		Poe Road (POE)	10.7	?	N	270	360	301.7	361.0	28.3	29.6	N/A	9.8
		Kimbloom Road (KRN)	17.8	?	N	270	360	99.1	136.2	12.5	33.7	N/A	4.4
		Westmorland Firestation (WST)	13.2	SL	F	90	180	205.4	233.9	27.2	29.7	8.7	14.8
24 Sierra Madre, CA, USA	28-Jun-91	El Centro (ELC)	13.6	SL	F	0	90	292.7	218.3	52.0	35.3	8.4	22.1
		Corralitos (COR)	2.3	SR	N	0	90	465.9	504.2	43.1	45.6	18.6	12.9
		Galvan College Physics Sci. Bldg. (GIL)	10.5	SL	N	337	63	293.4	378.4	31.5	27.6	12.4	7.0
		Gilroy 1 - Gavilan Coll. Water Tank (G01)	10.1	SR	N	0	90	412.4	406	38.7	29.1	14.5	7.9
		Lexington Dam - Left Abutment (LXD)	6.7	SR	F	0	90	443.1	387.4	131	119	43.6	25.5
		Los Gatos Presentation Center (LGP)	3.0	HR	F	0	90	633.8	426.1	102	59.2	54.9	36.6
		Saratoga - Aloha Avenue (STG)	8.3	SL	F	0	90	358.3	353.4	56.4	43.3	26.3	29.8
		Gilroy Historic Building (GHB)	11.1	?	N	90	180	270.4	145.9	32.6	30.2	11.1	14.1
		Capitola Fire Station (CAP)	15.8	SL	B	0	90	339.5	514.6	50.1	37.4	40.3	18.8
		Capitola Drive (BRN)	11.5	?	B	0	90	822.1	410.2	53.1	36.6	17.9	7.0
25 Erzincan, Turkey	23-Mar-92	Cogswell Dam (COG)	9.4	?	F	65	155	292.1	255.5	14.7	9.2	6.5	2.2
		Eaton Canyon Park - Aladena (ETN)	9.6	SL	F	0	90	354.3	311.9	22.1	16.4	4.3	2.4
		Mt. Wilson - Caltech Station (MTW)	11.9	HR	F	0	90	332.4	233.3	8.8	12.0	6.3	0.7
		San Marino - SW Academy (SNM)	15.6	SL	F	0	90	152	122.7	9.4	6.1	6.5	1.3
		Erzincan (ERZ)	2.0	SL	F	0	90	475.3	419.2	95.2	44.7	18.3	30.4
26 Joshua Tree, CA, USA	23-Apr-92	Fun Valley (FYI)	7.0	?	N	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		Cape Mendocino (CAP)	7.8	HR	F	0	90	1245.0	1404.0	738.9	58.8	119.0	60.3
		Petrolia (PGS)	8.9	SL	F	0	90	602.8	617.1	159.7	81.6	60.5	20.9
27 Petrolia, CA, USA	25-Apr-92	Fortuna (FOR)	13.5	SL or SR	N	0	90	122.3	114.2	47.9	21.2	29.6	5.9
		Rio Dell (RIO)	13.1	SL or SR	N	270	0	407.8	530.3	191.7	39.3	38.6	10.6
28 Landers, CA, USA	28-Jun-92	Joshua Tree - Fire Station (JSH)	11.6	SL	B	90	0	265.1	257.6	177.7	43.9	25.9	14.8
		Lucerne Valley (LUC)	1.1	SL	F	0	90	719.4	713.6	N/A	114.0	107.0	N/A
29 Northridge, CA, USA	17-Jan-94	Jensen Filtration Plant (JFA)	5.2	SL	F	292	22	385.6	604.3	105.0	94.3	34.1	45.1
		Pacoima Kugel Canyon (PKC)	7.4	HR	N	0	90	516.2	231.6	176.6	55.5	36.8	14.2
		Newhall: LA County Fire Station (NWH)	6.5	SL	F	0	90	637.9	537.3	118.0	49.3	30.7	28.3
		Arieta: Northhoff Fire Station (SFY)	8.0	SL	N	0	90	232.1	325.1	541.4	26.0	31.4	17.5
		Pacoima Dam - Downstream (PCD)	7.2	HR	F	175	265	488.8	240.2	180.4	48.5	18.9	16.1
		Rinaldi Receiving Station (RRS)	6.0	SL	F	228	318	869.7	382.3	835.9	173.0	61.5	50.7
		Symar Converter Station (SCG)	5.1	SL	F	52	142	576.5	640.7	770.8	134.0	103.0	40.8
		Symar Converter Station East (SCH)	5.0	SL	F	281	11	871.8	562.6	483.8	122.0	82.3	25.8
		Sequoyeda Veterans Hospital (VSP)	8.0	SL	F	270	360	708.1	742.7	458.5	65.5	73.1	33.2
		Los Angeles Dam (LDW)	5.6	SL	F	334	64	399.7	332.5	316.5	78.2	66.3	21.5
		Symar County Hospital (SYH)	5.5	SL	F	0	90	32	718.2	583.4	525.0	54.3	18.6
		Canoga Park Epiphany Church (CPC)	13.7	SL	N	106	196	345.3	349.9	409.6	52.9	36.7	13.9
		Canyon Country (CCY)	12.9	SL	F	0	270	476.9	369.2	279.9	48.7	45.8	18.5
		LA Fire Station (LF5)	19.2	SL	B	35	125	556.9	460.6	277.6	29.7	32.0	10.2
		LA Fire Station (LF6)	17.3	SL	B	279	9	385.5	403.6	312.8	56.4	77.0	19.4
		LA Westlake School (LWS)	19.9	SL	B	0	90	329.9	437.5	260.6	20.0	26.5	9.0
		LA Westlake School (LWS)	19.0	SL	B	0	90	197.8	274.4	158.2	13.7	28.0	7.0
		Newhall: Pico Canyon (NWS)	5.3	SL	F	316	46	404.1	320.3	280.5	117.0	73.8	29.0
		North Hollywood Coldwater Canyon (NHW)	11.3	SL	N	180	270	307.9	205.2	256.3	20.8	24.7	11.1
30 Northridge, CA, USA	17-Jan-94	Northridge White Oak Cov. Church (NRG)	11.5	SL	F	90	180	362.5	376.6	56.0	42.6	35.4	14.1
		Simi Valley (SMI)	12.4	SL	N	0	90	792.0	597.4	65.5	29.0	13.0	6.5

Table A2
Near-Fault Strong Ground Motion Database (Continued)

Location	Date	Near-Fault Station	Closest Distance * (km)	Site Code	Directivity	Original Horizontal Components (o)	Rotation Angle (o)	SN ** (cm/s ²)	Peak Acceleration SP ** (cm/s ²)	Vert (cm/s ²)	SN ** (cm/s)	Peak Velocity SP ** (cm/s)	Vert (cm/s)	SN ** (cm)	Peak Displacement SP ** (cm)	Vert (cm)	
30 Hanshin (Kobe), Japan	17-Jan-95	Sun Valley Grace Commun. Church (SVG)	9.3	SL	N	0	212	287.2	397.1	279.1	29.8	33.7	11.9	5.4	8.9	4.8	
		Tarzana Nursery (TAR)	14.8	SL or SR	B	0	212	1307.0	1959.0	1028.0	64.4	96.9	72.3	38.6	16.9	17.0	
		Kobe University (KBU)	0.7	HR	F	0	140	316.7	231.6	373.0	42.9	38.6	20.2	13.1	7.3	6.5	
		Kobe (KOB)	0.4	SL	F	0	140	838.0	537.9	336.2	95.7	53.4	38.3	24.5	10.3	10.3	
		Kobe Port Island, Surface (KPI)	3.2	SL	F	0	140	376.7	142.0	551.6	84.2	25.6	70.3	45.1	11.0	27.3	
		Takarazuka (TAZ)	0.4	SL	F	0	320	633.0	684.2	425.1	72.5	48.2	34.8	20.7	26.5	12.4	
		Takatori (TAK)	1.1	SL	F	0	140	669.0	593.6	266.5	169.0	62.8	16.0	45.0	22.9	4.5	
31 Aigion, Greece	15-Jun-95	Takatori (TAK)	1.1	SL	F	0	140	669.0	593.6	266.5	169.0	62.8	16.0	45.0	22.9	4.5	
		Nishi-Akashi (NIS)	10.5	SL	N	0	320	474.4	345.1	363.7	37.1	17.3	10.9	12.0	5.6	5.6	
		Aigion (AEG)	6.0	SL or SR	F	Long	-	490.0	535.0	193.0	40.9	52.1	17.0	8.2	9.0	4.7	
		Arcelik Arge Lab (ARC)	14.0	SR	F	0	90	131.1	207.4	81.7	44.3	16.8	8.2	65.4	67.2	7.5	
		Izmit (IZT)	4.8	SR	N	90	180	162.1	229.9	148.5	27.5	41.6	16.0	10.0	25.5	11.5	
		Sakarya (SKR)	3.1	SR	F	90	180	270	399.3	254.0	N/A	80.3	42.7	N/A	209.8	28.2	
		Duzce (DZC)	12.7	SL	F	270	180	315.0	362.9	478.2	41.4	51.9	19.5	25.6	17.4	9.6	
32 Izmit, Turkey	17-Aug-99	Yarimca-Pekim (YPT)	2.6	SL	F	90	0	316.1	225.7	236.5	86.8	87.8	31.5	149.8	184.6	43.6	
		Gebze (GBZ)	11.0	SR	F	270	0	256.8	142.7	193.0	41.4	28.7	11.2	33.6	32.0	6.6	
		CHY024	7.7	SL	F	0	270	276.4	162.1	141.4	51.1	42.8	47.1	94.8	34.6	28.9	
		CHY028	2.3	SL	N	0	270	626.5	749.9	336.7	63.5	84.2	30.7	24.8	27.1	22.4	
		CHY101	7.7	SL	F	0	270	333.0	390.5	162.3	65.9	108.4	27.3	54.0	84.0	24.0	
		TCU049	2.7	SL	F	0	90	270	273.4	240.9	178.0	57.1	59.5	27.2	128.0	121.0	21.9
		TCU051	6.9	SL	F	0	90	270	156.8	231.0	109.8	52.2	42.2	30.3	115.6	91.2	25.5
33 Chi-Chi, Taiwan	20-Sep-99	TCU052	0.8	SL	F	0	90	348.7	436.9	192.6	181.0	216.0	169.0	499.0	696.0	407.0	
		TCU053	4.6	SL	F	0	90	270	225.0	132.1	120.9	42.9	44.0	32.2	117.4	103.2	27.6
		TCU054	4.7	SL	F	0	90	270	143.3	190.4	133.0	45.9	45.1	29.8	121.4	124.0	28.5
		TCU055	6.5	SL	F	0	90	270	257.1	208.5	153.5	52.5	49.9	58.2	127.8	56.3	38.6
		TCU065	0.1	SL	F	0	90	270	772.5	563.3	258.9	132.0	90.0	68.8	183.0	134.0	59.5
		TCU067	0.2	SL	F	0	90	270	488.8	314.3	231.7	97.5	55.6	50.2	185.0	106.0	42.6
		TCU068	0.2	SL	F	0	90	501.5	364.0	519.4	281.0	291.0	229.0	710.0	857.0	451.0	
34 Duzce, Turkey	12-Nov-99	TCU071	4.1	SL	F	0	90	517.8	637.5	417.0	69.8	94.0	59.5	174.0	290.0	229.0	
		TCU072	6.8	SL	F	0	90	467.2	370.5	276.2	99.0	70.6	38.9	252.0	268.0	128.0	
		TCU074	11.4	SL	F	0	90	585.0	368.4	279.1	70.3	48.9	25.0	199.0	146.0	72.8	
		TCU075	0.6	SL	F	0	90	325.4	257.3	223.9	116.0	37.2	50.0	171.0	60.2	37.3	
		TCU076	2.3	SL	F	0	90	270	340.2	420.0	275.4	69.3	63.0	32.7	108.0	72.8	30.7
		TCU078	5.4	SL	F	0	90	438.2	302.4	172.2	43.2	32.3	18.8	123.0	108.0	36.5	
		TCU082	5.0	SL	F	0	90	270	222.3	182.5	130.5	52.1	43.1	35.4	150.0	97.8	29.5

HR, hard rock; SR, sedimentary and conglomerate rock; SL, soil and alluvium; SN, strike-normal; SP, strike-parallel; F, forward; N, neutral; B, backward; N/A, not available

* Normal distance from fault trace for events: 3, 6, 8, 11, 12, 13, 14, 16, 21, 22, 28, 30, 32, 34.

Normal distance from fault plane for events: 17, 18, 19, 20, 23, 24, 27, 29, 31, 33.

Distance based on references for events: 8, 9, 25.

Distance from point source (i.e., hypocentral distance) for events: 4, 5, 10.

** For Events 4, 5 and 10, rupture initiated at large depths producing strong pulses due to the directivity effect; these events have been modeled as point shear dislocation sources. Event 10 has been modeled as a line shear dislocation source and as a couple of point shear dislocation sources as well. The original horizontal components of these events have been rotated to the radial and transverse directions with respect to the epicenter instead of the strike-normal and strike-parallel orientations.

For Station C02 of the 1999 Chi-Chi earthquake, only a single horizontal ground motion component was recorded in the N55E direction. For historical reasons, this record has been included in the table assuming that the orientation of the horizontal component coincides with the strike-normal direction. It was essentially the first record where near-source effects were identified and successfully simulated (i.e., Housner and Trifunac, 1967; Aki, 1988).

For stations KOYNA, KAR, and AEG, the original horizontal components have not been rotated to the strike-normal and strike-parallel directions due to uncertainties regarding the strike direction of the causative fault plane or the precise orientation ground-motion components.

For the 1999 Izmit and 1999 Duzce, Turkey earthquakes, we have assumed that the NS and EW directions coincide with the strike-normal and strike-parallel directions, respectively. For the 1999 Chi-Chi, Taiwan earthquake, we have assumed that the NS and EW directions roughly coincide with the strike-parallel and strike-normal directions, respectively.